QUANTUM IAS & PCS

TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Indian Geography

PYQ-Based Revision Pointers

Chapters 1 to 26

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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Contents

NO.	CHAPTER	POINTERS	PAGE
	UNIT I · INDIA: POSITION, LAND AND COAST		
01	India: Location, Extent and Standard Time	26	4
02	Physical Divisions: Structure, Mountains, Peaks, Passes and Glaciers	59	6
03	Coastal Regions and Islands	19	9
04	States and Union Territories	17	11
05	Tribes of India	31	13
	UNIT II · DRAINAGE, CLIMATE AND NATURAL ENVIRONMENT		
06	Drainage I: The Himalayan Rivers	34	15
07	Drainage II: Peninsular Rivers and River Cities	35	18
08	Waterfalls, Lakes and Hot Springs	23	20
09	Climate and the Monsoon	23	22
10	Natural Disasters	15	24
11	Soils	25	26
12	Natural Vegetation and Forests	12	28
	UNIT III · WATER, AGRICULTURE AND LIVESTOCK		
13	Irrigation and Canals	13	30
14	Multipurpose River Valley Projects and Hydropower	45	31
15	Agriculture and the Green Revolution	30	33
16	Major Crops I: Crop Seasons, Wheat and Rice	37	35
17	Major Crops II: Cotton, Sugarcane, Oilseeds and Pulses	37	37
18	Silk, Plantation Crops and Spices	31	39
19	Farming Practices, Horticulture and Livestock	56	41
	UNIT IV · MINERALS, ENERGY AND INDUSTRY		
20	Mineral Resources I: Metallic and Non-metallic Minerals	55	43
21	Mineral Resources II: Coal, Petroleum and Natural Gas	54	45
22	Power and Energy Resources	37	48
23	Industries and Industrial Centres	39	50
24	Research Centres and Institutes	33	52
	UNIT V · TRANSPORT, TOURISM AND MISCELLANY		
25	Transport: Roads, Railways, Ports and Waterways	50	54
26	Tourism and Miscellaneous	18	56

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How to Use This Book

Every pointer is a real question. Each pointer is the answer to a UPPSC question asked between 1990 and 2025, written as a fact you can revise in seconds, with the related fact you need for the next variation.

Frequency marks. ■ and a red count (e.g. 4×) mark a fact asked 3 or more times. • and a gold count mark facts asked once or twice. The years follow; "one-liner" means a short starred question in the source with no year given.

Asked Most box. Each chapter opens with the facts UPPSC repeats most. Revise these first.

Diagrams. Flowcharts and river diagrams turn sequences (river courses, confluences, processes, orders from west to east) into one picture.

Trap Alerts. The places where the obvious answer is wrong or two facts are commonly swapped.

Last-Minute 10. The ten lines to read on the morning of the exam.

Pairs with the Masterbook. Use this book for revision after the TATTVA PYQ Masterbook, which has the full questions, options and explanations.

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Chapter 1

India: Location, Extent and Standard Time

Built from 26 UPPSC questions (1990–2022). Focus: Tropic of Cancer, IST and time sums, border states, boundary lines.

ASKED MOST · REVISE FIRST

- **Tropic of Cancer** crosses **8 states**: Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram; **not UP, not Manipur** 3× 1990, 2008, 2017
- Town **nearest the Tropic**: **Gandhinagar** (Jabalpur when Gandhinagar is absent) 4× 1999, 2002, 2010, 2017
- **IST = 82°30' E**; passes through **Mirzapur** near Prayagraj; 5 states (UP, MP, Chhattisgarh, Odisha, AP); not Maharashtra 3× 2006, 2010, 2013
- **Time sums**: 1° = 4 minutes, 15° = 1 hour; **east is ahead** 4× 1992, 2001, 2002, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

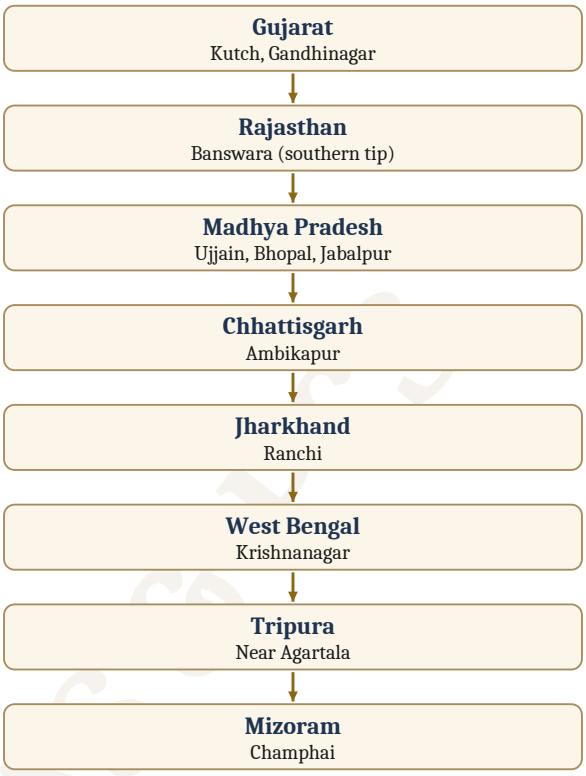
1.1 Size and Extent

- India is the **7th** largest country (not 5th or 6th) and has about **2.4%** of world land 2× 2006, 2022
- **Tropic of Cancer** divides India into two nearly equal halves; India is **not wholly** in the torrid zone 2× 2006, 2022
- India is now the **most populous** country (passed China, April 2023) 1× 2016
- Mainland: **8°4' N to 37°6' N, 68°7' E to 97°25' E**; area **32.87 lakh sq km**

1.2 The Tropic of Cancer

- States crossed: **Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, WB, Tripura, Mizoram** (west to east) 2× 1990, 2017
- **Manipur** lies wholly **north** of the Tropic 1× 2008
- Of Tripura, Mizoram, WB and Manipur, the Tropic does **not** cross **Manipur** 1× one-liner
- Nearest to the Tropic (23°26' N): **Gandhinagar** {1999, 2002, 2010}; next option set: **Jabalpur** 1× 2017
- **Mahi** is the river that crosses the Tropic **twice** 1× 2016

TROPIC OF CANCER, WEST TO EAST (8 STATES)



TRAP ALERT

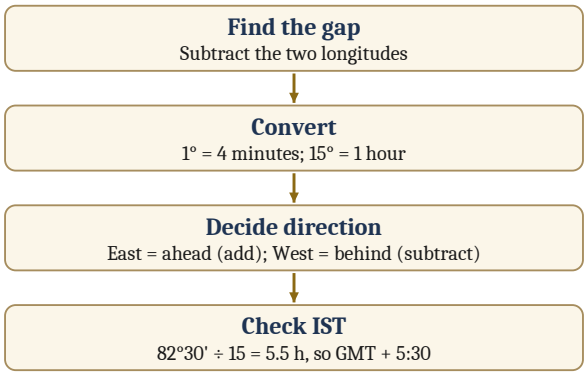
- **UP never touches the Tropic** (its southern tip is about 23°52' N). Strike out UP first.
- "Nearest town" answers change with the option set: Gandhinagar first, then Jabalpur.

1.3 Longitude and Indian Standard Time

- IST meridian is **82°30' E** (GMT + 5:30) 1× 2013
- It does **not** pass through **Maharashtra** 1× 2010
- IST is taken near **Prayagraj (Mirzapur)** 1× one-liner
- Gujarat to Arunachal (Walong): about 29° of longitude = about **2 hours** 1× 1992
- Noon at IST, local time at 120° E: 37°30' × 4 = 150 min ahead = **14:30** 1× 2001
- Midnight at IST, 6 a.m. at a place: 6 h = 90° east, so **172°30' E** 1× 2002
- Sunrise 5 a.m. at **Tirap**: Kandla sees it about **7 a.m.** 1× 2010

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HOW TO SOLVE ANY TIME QUESTION



1.4 Extreme Points

- Southernmost point of **India**: **Indira Point**, **Great Nicobar** **2×** 2002, one-liner
- Southernmost point of the **mainland**: **Kanyakumari**

Direction	Point	State / UT
North	Indira Col (Siachen)	Ladakh
South	Indira Point	Andaman and Nicobar
South (mainland)	Kanyakumari	Tamil Nadu
East	Kibithu / Dong	Arunachal Pradesh
West	Guhar Moti (Sir Creek)	Gujarat

1.5 Neighbours and Border States

- **Manipur** does **not** border Bangladesh **2×** 2002, 2004
- **Nagaland** does **not** border Bangladesh **1×** one-liner
- **Assam** does **not** border Myanmar **1×** 2012
- Border with Pakistan: **Gujarat, Rajasthan, Punjab, J&K** (now UTs J&K and Ladakh) **2×** 1990, 2001
- **Tripura** has an international border on **three sides** (Bangladesh) **1×** 2008

Neighbour	Border (km)	Indian states / UTs
Bangladesh	4,097 (longest)	WB, Assam, Meghalaya, Tripura, Mizoram
China	3,488	Ladakh, HP, Uttarakhand, Sikkim, Arunachal
Pakistan	3,323	Gujarat, Rajasthan, Punjab, J&K, Ladakh
Nepal	1,751	Uttarakhand, **UP**, Bihar, WB, Sikkim
Myanmar	1,643	Arunachal, Nagaland, Manipur, Mizoram
Bhutan	699	Sikkim, WB, Assam, Arunachal
Afghanistan	106 (shortest)	Ladakh (PoK area)

UP LINK

UP borders only **Nepal**, through 7 districts: Pilibhit, Lakhimpur Kheri, Bahraich, Shravasti, Balrampur, Siddharthnagar, Maharajganj. The IST meridian passes through **Mirzapur**.

1.6 Boundary Lines

- India-Pakistan boundary: **Radcliffe Line** (1947) **1×** 1996
- **Durand Line**: with **Afghanistan** (1893) **1×** 2006
- **McMahon Line**: **India-China** (1914, Simla) **1×** 2018

Line	Year	Between
Radcliffe	1947	India-Pakistan, India-Bangladesh
Durand	1893	British India-Afghanistan
McMahon	1914	India-China (Arunachal)
LoC	1972	India-Pakistan (J&K)
Maginot	1930s	France-Germany (never Indian)

LAST-MINUTE 10

1. India: **7th** largest, **2.4%** of land, **most populous** since 2023
2. Tropic of Cancer: **8 states**; not UP, not Manipur
3. **Gandhinagar** nearest the Tropic; **Mahi** crosses it twice
4. IST **82°30' E**, **Mirzapur**; 5 states; not Maharashtra
5. **1° = 4 min**; east is ahead
6. Southernmost: **Indira Point**; mainland: **Kanyakumari**
7. Longest border **Bangladesh**; shortest **Afghanistan**
8. **Tripura**: Bangladesh on three sides
9. Assam does not touch Myanmar; Manipur does not touch Bangladesh
10. **Radcliffe**-Pakistan, **Durand**-Afghanistan, **McMahon**-China

Chapter 2

Physical Divisions: Structure, Mountains, Peaks, Passes and Glaciers

Built from 77 UPPSC questions (1990–2025), the biggest geography chapter. Focus: passes and their states, peaks, hills, Himalayan ranges, glaciers.

ASKED MOST · REVISE FIRST

- **Passes and states** are asked almost every year: **Nathu La, Jelep La = Sikkim**; **Bomdi La, Se La, Bum La = Arunachal**; **Lipulekh, Niti, Mana = Uttarakhand**; **Shipki La, Rohtang = HP**; **Zoji La = Ladakh** 13× 1998, 2001, 2004, 2006, 2008, 2010, 2011, 2013, 2014, 2015, 2016, 2017, 2023
- **Highest peak of each state** (match): Anaimudi-Kerala, Doddabetta-TN, Saramati-Nagaland, Guru Shikhar-Rajasthan, Dhupgarh-MP, Nanda Devi-Uttarakhand 3× 2018, 2021, 2025
- **Anaimudi** (2,695 m) = highest peak of **South India / the Peninsula** 4× 2005, 2012, 2013, one-liner
- **Aravalli** = **oldest** and a **residual** mountain 4× 1991, 2005, 2010, 2017
- Largest glacier: **Siachen** (Gangotri if Siachen is not offered) 3× 1995, 2013, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 Geology and Rock Systems

- Palaeomagnetism shows the Indian landmass moved **northward** 1× 1998
- The **western coast** is not depositional; it is a **faulted (emergent-submergent)** coast 1× 2004
- Latest of the old rock systems: **Gondwana** (98% of India's coal) 1× 2016
- Deccan Trap intertrappean beds have **freshwater** (not sea) fossils 1× 2024

ROCK SYSTEMS, OLDEST TO YOUNGEST

Archaean
Gneiss, schist; no fossils

↓

Dharwar
Metals: iron, gold (Kolar), manganese

↓

Cuddapah, Vindhyan
Sandstone (Red Fort), Panna diamonds

↓

Gondwana
About 98% of India's coal

↓

Deccan Trap
Basalt; black soil

↓

Tertiary, Quaternary
Himalaya, petroleum; alluvial plains

2.2 The Himalaya

- Himalaya is **still rising**: the Indian plate is pushing into the Eurasian plate (A and R true, R explains A) 1× 2024
- Most seismic zones are in the Himalaya because of **longitudinal thrusts** 1× 2019
- **Lesser Himalaya**: marine fossils; **Shiwalik**: remains of early humans; Greater Himalaya sediments are not fossil-less 1× 2019
- **Himachal** = the **Lesser Himalaya** 1× 2008
- Youngest range: **Shiwalik** {2010}; if only Himalayan vs Peninsular ranges are offered: **Himadri** 1× 2020
- **Kumaon Himalaya** lies between the **Satluj and Kali** 1× 2016

HIMALAYAN RANGES, NORTH TO SOUTH

Trans-Himalaya
Karakoram, Ladakh, Zaskar; K2

↓

Greater Himalaya (Himadri)
Highest peaks, about 6,100 m

↓

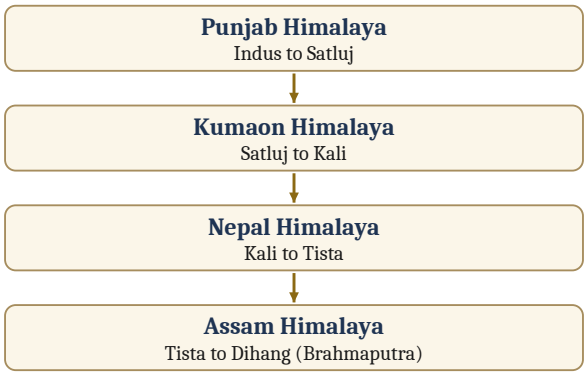
Lesser Himalaya (Himachal)
Pir Panjal, Dhauladhar; hill stations

↓

Shiwalik (Outer)
Youngest; duns such as Dehradun

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REGIONAL DIVISIONS, WEST TO EAST (BURRARD)



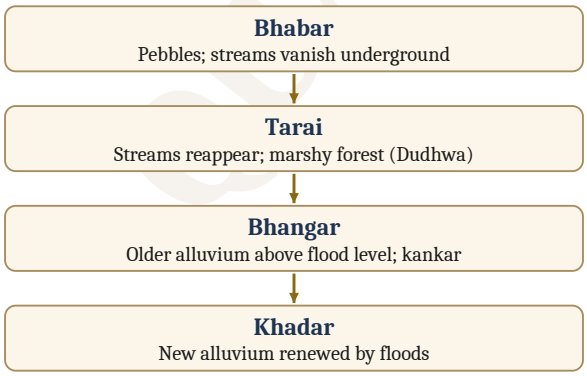
2.3 Ranges, Valleys and Tunnels

- **Pir Panjal** range is in **Jammu and Kashmir** 1× 2007
- **Atal Tunnel** (Rohtang): BRO, Manali to Lahaul-Spiti, under the **Pir Panjal** 1× 2016
- Atal Tunnel is the longest highway tunnel **above 10,000 ft**, not the world's longest (only statement 2 true) 1× 2025
- **Kashmir valley**: between the **Pir Panjal and Great Himalaya** 1× 2020
- **Kullu valley**: between the **Dhauladhar and Pir Panjal** 2× 1999, 2002
- **Nelang valley** is in **Uttarakhand** (Uttarkashi) 1× 2016
- **Imphal basin** is a **lacustrine plain** 1× 2019
- **Aksai Chin** is part of the **Ladakh plateau** 1× 1995
- **Shaksgam valley**: PoK area given by Pakistan to China (1963); India protests China's construction 1× 2024
- **Tripura** does not lie along the **Patkai** hills 1× 2015

2.4 The Northern Plains

- Gravelly plain at the foot of the Shiwalik: **Bhabar** {2007}; it is a **piedmont** region 1× 1994
- **Kuttanad** (Kerala) is India's **lowest** region, below sea level 1× 2015

NORTHERN PLAIN BELTS, FROM THE HILLS SOUTHWARD



2.5 Old Mountains and Peninsular Hills

- **Aravalli** = oldest mountains {1991, 2010, 2017}; a **residual** mountain {2005}; mainly in **Rajasthan** 1× one-liner
- **Satpura** lies between the **Narmada and Tapti** 1× 2007
- Central India hills, **west to east**: **Satpura, Mahadeo, Maikal, Chotanagpur** 1× 2019
- **Nilgiri** hills: junction of **Karnataka, Kerala and Tamil Nadu** 1× 2015
- **Cardamom hills** (**Yela Mala**): **Kerala-Tamil Nadu** border 3× 2008, 2010, 2023
- **Mandav hills** are in **Gujarat**, not Maharashtra 2× 2011, 2011
- Hills match: **Garo-Meghalaya, Miri-Arunachal, Kollimalai-Tamil Nadu, Dalma-Jharkhand** 1× 2017
- Hills match: **Bateshwar-Bihar, Bilari-Karnataka, Chin-Manipur, Dhosi-Haryana** 1× 2022
- **Western Ghats**: a **block mountain**, gentle eastern slope, **basalt** in the north 1× 2017
- **Kodaikanal** is in the **Palani** hills, not the Anaimalai 1× 2013

2.6 Peaks

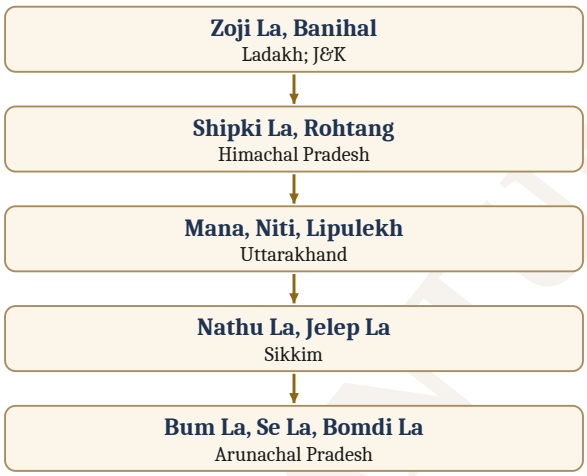
- Highest peak of the **Peninsula**: **Anaimudi** 4× 2005, 2012, 2013, one-liner
- Highest peak claimed by India: **K2** (PoK); highest in Indian control: **Kanchenjunga**; wholly in India: **Nanda Devi** 2× 1990, 2009
- Peaks east to west: **Namcha Barwa, Kanchenjunga, Everest, Nanda Devi** 1× 2008
- **Cho Oyu, Lhotse** are Himalayan; Annamalai, Sirumalai are not 1× 2022
- Not in India: **Gosainthan** (Tibet) {2005}; **Gurla Mandhata, Namcha Barwa** 1× 2019
- **Trishul** peak: Uttarakhand (Garhwal) 1× 2021
- State peaks: **Kerala-Anaimudi, Nagaland-Saramati, Uttarakhand-Nanda Devi, TN-Doddabetta** 1× 2018
- State peaks: **TN-Doddabetta, Rajasthan-Guru Shikhar, Nagaland-Saramati, MP-Dhupgarh** 2× 2021, 2025
- First Indian woman on Everest: **Bachendri Pal** (1984) 1× 1996

State	Highest peak (m)
Sikkim	Kanchenjunga (8,586)
Uttarakhand	Nanda Devi (7,816)
Himachal	Reo Purgyil (6,816)
Nagaland	Saramati (3,826)
Kerala	Anaimudi (2,695)
Tamil Nadu	Doddabetta (2,637)
Karnataka	Mullayanagiri (1,930)
Rajasthan	Guru Shikhar (1,722)
Odisha	Deomali (1,672)
MP / Jharkhand	Dhupgarh / Parasnath (1,350)

2.7 Mountain Passes

- **Se La** is in **Arunachal**, not Uttarakhand 1× 2015
- **Bhor Ghat** is in **Maharashtra**, not HP 1× 1998
- Western Ghat passes north to south: **Thal Ghat, Bhor Ghat, Pal Ghat** 1× 2023
- **Khyber** is not a Western Ghats pass (it is in Pakistan-Afghanistan) 1× 2016
- **Niti** = **Uttarakhand** {2023}; **Lipulekh** = **Uttarakhand** (not UP or Ladakh) 4× 2001, 2008, 2014, one-liner
- **Nathu La** = **Sikkim** (not Meghalaya or Arunachal) 4× 2010, 2013, 2017, one-liner
- **Zoji La** leads to **Leh** (now in Ladakh, not Arunachal) 2× 2016, 2023
- **Rohtang** = **Himachal Pradesh** 1× one-liner
- Match: **Banihal-J&K, Nathu La-Sikkim, Niti-Uttarakhand, Shipki-HP** 2× 2004, 2011
- Match: **Bum La-Arunachal, Jelep La-Sikkim, Muling La-Uttarakhand, Shipki La-HP** 1× 2006
- Match: **Mana-Uttarakhand, Nathu La-Sikkim, Zoji La-J&K (Ladakh), Shipki La-HP** 1× 2016
- Heights: **Debsa 5,360, Bara-lacha 4,850, Rohtang 3,978, Banihal 2,832 m** 1× 2024

HIMALAYAN PASSES, WEST TO EAST



TRAP ALERT

- Order asked in 2009: **Shipki La, Lipulekh, Nathu La, Bomdi La** (west to east).
- **Bomdila** is the town; the pass is **Bomdi La**. **Jelep La** and Nathu La both lead to the Chumbi valley.

2.8 Glaciers

- Largest: **Siachen** {2013, 2017}; without Siachen in the options: **Gangotri** 1× 1995
- **Chaurabari** glacier is **north of Kedarnath** 2× 2004, 2010
- **Milam** glacier is in **Kumaon** (Pithoragarh) 1× 2014

Glacier	Where	Feeds
Siachen (76 km)	Karakoram, Ladakh	Nubra
Gangotri (30 km)	Uttarkashi	Bhagirathi
Zemu	Sikkim	Tista
Milam, Pindari	Kumaon	Gori Ganga, Pindar
Chaurabari	Kedarnath	Mandakini

2.9 The Peninsular Plateau

- **Meghalaya plateau** is part of the **Peninsular plateau** 1× 2013
- **Dhanbad** is densely populated because of **mining and industry** 1× 2007
- **Dandakaranya** lies in **Chhattisgarh and Odisha** 1× 2008

UP LINK

Bundelkhand plateau (Jhansi, Banda, Chitrakoot) is the Peninsular part of UP. The **Tarai** of Pilibhit, Lakhimpur Kheri and Bahraich holds **Dudhwa National Park**. UP has no high peaks or passes; **Lipulekh** is often wrongly given as UP but is in Uttarakhand.

LAST-MINUTE 10

1. **Nathu La, Jelep La** = Sikkim; **Se La, Bum La, Bomdi La** = Arunachal
2. **Lipulekh, Niti, Mana** = Uttarakhand; **Shipki La, Rohtang** = HP; **Zoji La** = Ladakh
3. **Anaimudi** = highest in the Peninsula; **Doddabetta** = Tamil Nadu
4. **Aravalli** oldest; **Shiwalik** youngest
5. **Kumaon** Himalaya: Satluj to Kali
6. **Kullu**: Dhauladhar-Pir Panjal; **Kashmir**: Pir Panjal-Great Himalaya
7. **Bhabar** at the foot of the hills, then Tarai
8. **Satpura** between Narmada and Tapti; **Mandav** in Gujarat
9. **Siachen** largest glacier; **Chaurabari** above Kedarnath
10. **Gondwana** = coal; Kuttanad = lowest land

Chapter 3

Coastal Regions and Islands

Built from 21 UPPSC questions (2001–2016). Focus: longest coastline, named coasts, island channels, Barren and Lakshadweep.

ASKED MOST · REVISE FIRST

- ■ **Gujarat** has the **longest coastline** 4× 2006, 2007, 2016, 2018
- • **Ten Degree Channel** separates the **Andaman** group from the **Nicobar** group 2× 2005, one-liner
- • **Barren Island** (Andaman Sea, Bay of Bengal) is India's only **active volcano** 2× 2010, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Sea Level and Maritime Zones

- India's **mean sea level** is measured at **Chennai** 1× 2013
- **Territorial waters**: **12 nautical miles** from the coast 1× 2005
- ■ Longest coastline: **Gujarat** (about 2,340 km) 4× 2006, 2007, 2016, 2018
- • Coastal states: **9** (plus 4 coastal UTs) 1× one-liner

MARITIME ZONES, FROM THE COAST OUTWARD

Territorial sea
12 nautical miles; full sovereignty

↓

Contiguous zone
24 nm; customs, immigration

↓

Exclusive Economic Zone
200 nm; fishing, mining, drilling

↓

Continental shelf
To the margin edge (up to 350 nm)

TRAP ALERT

- Coastline length changed: old figure **7,516.6 km**, new count **11,098.81 km**. Read the options. **Goa** has the shortest state coastline.

3.2 Named Coasts and Coastal Towns

- **Coromandel** coast: **Krishna delta to Cape Comorin** 1× 2015
- Northern part of the western coastal plain: **Konkan** 1× 2016
- West-coast towns north to south: **Janjira, Sindhudurg, Kannur, Nagercoil** 1× 2001
- West-coast towns among the options: **Janjira, Udupi** (Auroville and Tuticorin are on the east coast) 1× 2002
- Beaches: **Digha-WB, Gopalpur-Odisha, Calangute-Goa, Marina-Tamil Nadu** 1× 2008

WEST COAST, NORTH TO SOUTH

Kathiawar coast
Gujarat: gulfs of Kutch, Khambhat

↓

Konkan
Daman to Goa; Mumbai

↓

Kannada coast
Goa to Mangaluru

↓

Malabar
Mangaluru to Kanyakumari; backwaters (kayals)

EAST COAST, NORTH TO SOUTH

Utkal coast
Odisha; Chilika lake

↓

Northern Circars
Mahanadi to Krishna

↓

Coromandel
Krishna delta to Kanyakumari; Pulicat

3.3 Andaman and Nicobar Islands

- Highest peak: **Saddle Peak** (North Andaman) 1× 2009
- **Ten Degree Channel**: **Andaman** from **Nicobar** 2× 2005, one-liner
- **Barren Island** is in the **Bay of Bengal** (Andaman Sea); India's **volcanic** island 2× 2010, one-liner

Channel	Separates
Ten Degree	Andaman group from Nicobar group
Duncan Passage	South Andaman from Little Andaman
Coco Channel	North Andaman from Coco Is. (Myanmar)
Nine Degree	Minicoy from main Lakshadweep
Eight Degree	Minicoy from the Maldives
Palk Strait	Tamil Nadu from Sri Lanka

3.4 Lakshadweep and Other Islands

- **Lakshadweep** is a **coral** island group 1× 2013
- **Minicoy** is one of the remotest islands from the Indian coast 1× 2013
- **Rameswaram** lies between India and Sri Lanka 1× 2006
- **Adam's Bridge** begins at **Dhanushkodi** 1× 2010
- Most populous island: **Salsette** (Mumbai) 1× 2016
- Match: **Viyant Syodhar-Gulf of Kutch, Piram-Gulf of Khambhat, Dwarka-Arabian Sea coast, Diu-Kathiawar coast** 1× 2010
- **Kori Creek** is in the **Rann of Kutch** 2× 2008, 2011

TRAP ALERT

- **Ten Degree** = Andaman-Nicobar. **Nine and Eight Degree** belong to the Lakshadweep side. Barren is **active**; Narcondam is **dormant**.

UP LINK

UP is landlocked, but UPPSC asks island and coast facts as standard map knowledge. Remember **Salsette** (Mumbai), **Majuli** (largest river island) and **Sriharikota** (ISRO launch site).

LAST-MINUTE 10

1. **Gujarat**: longest coastline; **Goa**: shortest
2. Mean sea level measured at **Chennai**
3. Territorial sea **12 nm**; EEZ **200 nm**
4. **Konkan** = north of the west coast; **Coromandel** = Krishna to Kanyakumari
5. Janjira, Sindhudurg, Kannur, Nagercoil (north to south)
6. **Saddle Peak** = highest in Andaman and Nicobar
7. **Ten Degree Channel** = Andaman | Nicobar
8. **Barren** = active volcano; **Lakshadweep** = coral
9. **Salsette** = most populous island; **Minicoy** = remotest
10. Adam's Bridge from **Dhanushkodi**; **Kori Creek** in Kutch

QUANTUM IAS & PCS

Chapter 4

States and Union Territories

Built from 18 UPPSC questions (1990–2019). Focus: area ranks, shared borders, capitals, NCR, Union Territories.

ASKED MOST · REVISE FIRST

- Area ranks: **Rajasthan > MP > Maharashtra > UP**; UP is **4th** by area **3×** 2016, 2019, one-liner
- **NCR** includes parts of **Haryana, UP and Rajasthan** only; not Punjab or MP **3×** 2011, 2017, one-liner
- **Sonbhadra** touches **four states** (MP, Chhattisgarh, Jharkhand, Bihar) **1×** 2015

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

4.1 Places and their States

- **Palanpur** is in **Gujarat** **1×** 2004
- **Malnad** belongs to **Karnataka**, not Andhra Pradesh **1×** 2000

4.2 Size, Population and Districts

- UP is **4th** by area **1×** 2016
- About **30%** of India's area lies in **Rajasthan, MP and Maharashtra** **1×** one-liner
- **Maharashtra**: **2nd** in population, **3rd** in area (2011) **1×** 2019
- Most districts: **Uttar Pradesh** (75) **1×** 2006
- **Thar**: densely populated desert, about 10,000 years old, spreading partly through human misuse; less grazing land as farming spreads (all four true) **1×** 2008

Rank	By area	By population (2011)
1	Rajasthan	Uttar Pradesh
2	Madhya Pradesh	Maharashtra
3	Maharashtra	Bihar
4	**Uttar Pradesh**	West Bengal
Smallest	Goa	Sikkim

4.3 Shared Borders

- In 1993 undivided MP touched **7 states**; today MP touches **5** **1×** 1993
- **Chhattisgarh** does **not** border **Bihar** **1×** 2013
- **Sonbhadra** touches the boundaries of **four states** **1×** 2015
- **Telangana** borders **AP, Karnataka, Maharashtra, Chhattisgarh** **1×** 2015

UP LINK

UP borders **8 states + Delhi**, the most of any state: Uttarakhand, **Himachal** (Saharanpur), Haryana, Rajasthan, MP, **Chhattisgarh** and Jharkhand (both via Sonbhadra), Bihar. It does **not** border Punjab or Gujarat. It has **75 districts**, the most in India.

4.4 Capitals and the NCR

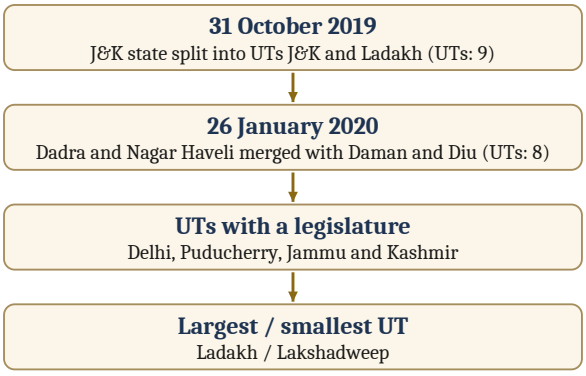
- **Manipur-Aizawl** is wrong; Manipur's capital is **Imphal** (Aizawl is Mizoram's) **1×** 1990
- **Assam-Dispur, Nagaland-Kohima, Arunachal-Itanagar, Meghalaya-Shillong** **1×** 1994
- NCR towns among the options: **Khurja, Karnal, Rohtak** (not Ambala) **1×** 2011
- NCR does **not** include **Punjab** {?} or **Madhya Pradesh** **1×** 2017

North-east state	Capital
Assam	Dispur
Arunachal Pradesh	Itanagar
Meghalaya	Shillong
Nagaland	Kohima
Manipur	Imphal
Mizoram	Aizawl
Tripura	Agartala
Sikkim	Gangtok

4.5 Union Territories

- Largest UT among the options in 2014: **Delhi**; now **Ladakh** is the largest UT **1×** 2014
- **Puducherry** is spread over **Tamil Nadu, Kerala and Andhra Pradesh** (Puducherry, Karaikal, Mahe, Yanam) **1×** 2006

UNION TERRITORIES SINCE 2019 (NOW 8)



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TRAP ALERT

- The NCR question repeats with different wrong options: **Punjab**, **Himachal**, **MP**. Only Haryana, UP and Rajasthan are in the NCR.
- Older keys call **Delhi** the largest UT; after 2019 it is **Ladakh**.

LAST-MINUTE 10

1. Area: Rajasthan, MP, Maharashtra, **UP (4th)**
2. Population: **UP** first; Maharashtra 2nd (and 3rd by area)
3. UP: most districts (75); borders 8 states + Delhi
4. **Sonbhadra** touches 4 states
5. Chhattisgarh does not touch Bihar
6. **Imphal** (Manipur), **Aizawl** (Mizoram), **Dispur** (Assam)
7. NCR = Haryana, UP, Rajasthan
8. **Ladakh** largest UT; 8 UTs today
9. Puducherry in TN, Kerala, AP
10. **Malnad** = Karnataka; **Palanpur** = Gujarat

QUANTUM IAS & PCS

Chapter 5

Tribes of India

Built from 35 UPPSC questions (1991–2022). Focus: tribe-state matching, the Tharu and Bhotia, Toda polyandry, Andaman tribes.

ASKED MOST · REVISE FIRST

- **Tribe-state matching** is asked almost every year: **Toda = Tamil Nadu (Nilgiris), Lepcha = Sikkim, Birhor/Santhal = Jharkhand, Gaddi = Himachal, Jarawa = Andaman** 10× 2002, 2002, 2013, 2014, 2015, 2016, 2017, 2019, 2020, 2022
- **Tharu** live in **Uttarakhand and UP**; largest ST of Uttarakhand; **Diwali is mourning** for them 6× 1991, 2005, 2005, 2015, 2017, one-liner
- **Toda** (Nilgiri hills) and **Jaunsari** practise **polyandry** 5× 1993, 2005, 2006, 2009, 2013
- **Gaddi** = tribe of the **Dhauladhar** range (Himachal) 3× 2001, 2002, 2003
- **Jarawa** live in the **Andaman and Nicobar Islands** 3× 2013, 2016, one-liner

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

5.1 Racial Groups and Identity

- North-east hill tribes are **Mongoloid** 1× 2003
- **Santhal** are **Proto-Australoid** 1× 2002
- Tribes are identified by **distinct culture and separate habitat** 1× 1992

Racial group	Example tribes
Negrito	Onge, Jarawa, Sentinelese, Great Andamanese
Proto-Australoid	Santhal, Munda, Ho, Oraon, Bhil
Mongoloid	Naga, Khasi, Garo, Lepcha, Bhutia

5.2 Tribes of UP and Uttarakhand

- **Tharu**: **Uttarakhand and Uttar Pradesh** 2× 1991, 2017
- ■ **Diwali** is a festival of **mourning** for the **Tharu** 3× 2005, 2015, one-liner
- Largest ST of Uttarakhand: **Tharu** 1× 2005
- **Transhumance** is practised by the **Bhotia** 1× 1997
- "Bhotia-UP" pairs have **no wrong option** (Bhotia are notified in UP too), so the source marks them (*) 2× 2004, 2010

TRAP ALERT

- **Bhotia** (UP, Uttarakhand) and **Bhutia** (Sikkim) are **different** tribes. A "Bhutia" option in a Sikkim pair is correct; a "Bhotia-UP" pair is also correct.

5.3 Customs and Festivals

- **Polyandry**: **Toda** {2005}; **Jaunsari and Toda** (not Gond or Naga) 2× 2006, 2009
- **Jhum** (shifting cultivation) is practised by the **Khasi** 1× 2010
- **Sarhul** (sal-blossom festival): **Oraon, Munda, Santhal** of Jharkhand 1× 2005
- **Lohasur** (iron god) is worshipped by the **Agaria** (iron smelters) 1× 2016
- **Onam-Andhra Pradesh** is wrong; Onam is **Kerala's** 1× 2005
- **Sikkim-Tibetan** is wrong as the main language; it is **Nepali** 1× 2006
- **Khasi and Garo** speakers live in **Meghalaya** 1× 2002

Shifting cultivation name	Region
Jhum	North-east
Podu	Andhra Pradesh
Bewar, Dahiya	Madhya Pradesh
Penda, Dippa	Chhattisgarh (Bastar)
Kumari	Western Ghats
Milpa / Ladang	Mexico / Indonesia

5.4 Tribes and their States

- Largest tribe of India: **Bhil** 1× 2009
- **Jarawa-Andaman, Kuki-Manipur, Tharu-UP, Gond-MP** 1× 2019
- Correct pairs: **Keria-Odisha, Paliyan-Tamil Nadu** (not Kuki-UP, Yanadi-Rajasthan) 1× 2022
- **Bhil-Rajasthan, Santhal-Jharkhand, Raji-Uttarakhand, Lepcha-Sikkim** 1× 2002
- **Jaunsari-Uttarakhand, Santhal-Jharkhand, Bhil-MP, Lepcha-Sikkim** 1× 2002
- **Birhor-Jharkhand, Bhutia-Sikkim, Toda-Tamil Nadu, Sentinelese-Andaman** 2× 2013, 2014
- **Toda-Tamil Nadu, Lepcha-Sikkim, Birhor-Jharkhand, Garo-Meghalaya** 1× 2015
- **Gaddi-Himachal, Toda-Tamil Nadu, Khasi-Meghalaya, Naga-Manipur** 1× 2017
- **Bhil-Rajasthan, Gond-Chhattisgarh, Limboo-Sikkim, Konyak-Nagaland** 1× 2016
- **Khas-Arunachal** is wrong; Khas live in the **western Himalaya** 1× 2020
- **Toda** live in the **Nilgiri hills** 2× 1993, 2013
- ■ **Gaddi**: **Dhauladhar** range 3× 2001, 2002, 2003
- **Tamaria**: **Jharkhand** 1× 2021
- None of Chenchu, Lepcha, Dafla, Dafar is a Kerala tribe (marked *) 1× 2013

Tribe	State
Toda, Kota, Badaga, Paliyan	Tamil Nadu

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Tribe	State
Chenchu, Yanadi	Andhra Pradesh
Gaddi	Himachal Pradesh
Jaunsari, Tharu, Bhotia, Raji	Uttarakhand
Lepcha, Bhutia, Limboo	Sikkim
Konyak, Ao, Angami, Rengma	Nagaland
Khasi, Garo, Jaintia	Meghalaya
Apatani, Adi, Nyishi (Dafla)	Arunachal
Kuki	Manipur
Santhal, Munda, Ho, Birhor	Jharkhand
Bhil	Rajasthan, MP, Gujarat
Gond, Baiga, Agaria	MP, Chhattisgarh

5.5 Andaman and Nicobar Tribes

- **Jarawa** live in the **Andaman and Nicobar Islands** {2013, 2016}; called India's **most primitive** tribe 1× one-liner
- **Rengma** is a **Naga** tribe, not native to the islands 1× 2020

ISLAND TRIBES, NORTH TO SOUTH



UP LINK

UP's main tribes are the **Tharu** (Tarai: Lakhimpur Kheri, Bahraich, Balrampur), **Buksa** and **Bhotia**. The Tharu mourn on **Diwali**. **Sonbhadra** has Gond, Kharwar and Cheru communities.

LAST-MINUTE 10

1. **Bhil** = largest tribe
2. **Toda** = Nilgiris, polyandry; **Jaunsari** = polyandry
3. **Tharu** = UP and Uttarakhand; Diwali mourning
4. **Bhotia** = transhumance; not the same as Bhutia (Sikkim)
5. **Gaddi** = Dhauladhar (HP)
6. **Lepcha** = Sikkim; **Birhor, Santhal** = Jharkhand
7. **Khasi, Garo** = Meghalaya; Khasi practise jhum
8. **Sarhul** = Oraon, Munda; **Agaria** worship Lohasur
9. **Jarawa, Onge, Sentinelese** = Andaman (Negrito)
10. **Rengma** = Naga, not Andaman

Chapter 6

Drainage I: The Himalayan Rivers

Built from 36 UPPSC questions (1990–2025). Focus: the Ganga system, Satluj, Bundelkhand rivers, Brahmaputra and Indus.

ASKED MOST · REVISE FIRST

- **Satluj** is the most-asked river: rises at **Rakas Tal** (Tibet), enters India at **Shipki La**, cuts across **all Himalayan ranges**, is **trans-Himalayan** and **does not rise in India** 6× 2010, 2012, 2013, 2014, 2020, 2023
- **Devprayag** = **Alaknanda + Bhagirathi**; the river is called **Ganga** from here 4× 2002, 2016, 2023, 2024
- **Betwa** (Bundelkhand): joins the **Yamuna**; **Rajghat dam** is on it; **Ken-Betwa** (UP-MP) is India's first river link 4× 2006, 2006, 2014, 2015
- Himalayan rivers are **perennial** because of **snow and glacier melt**, not the monsoon (Assertion-Reason) 2× 2011, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Nature of Himalayan Rivers

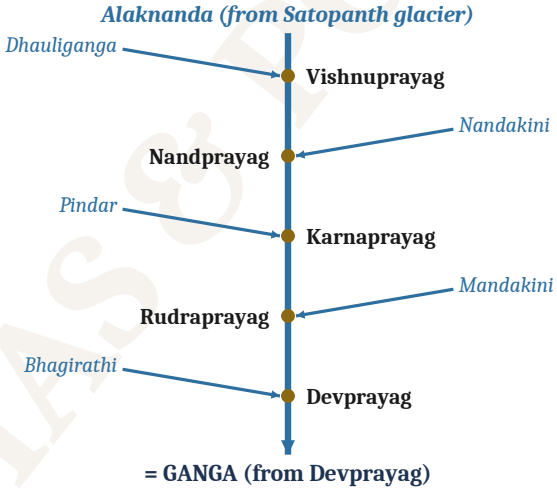
- Himalayan rivers are **perennial**: summer **snow-melt** keeps them flowing; the monsoon explains only the rainy-season flood 2× 2011, 2025
- **Ganga** is an **antecedent** river: older than the mountains, it kept cutting down as they rose 1× 2013
- **Chambal** is a **superimposed** river (also Son, Banas, Subarnarekha) 1× 2007
- **Satluj** cuts across **all ranges** of the Himalaya (Zaskar, Greater, Lesser, Shiwalik) 2× 2012, 2014
- **Trans-Himalayan** rivers (rise north of the Himalaya): **Satluj**, Indus, Brahmaputra 1× 2020

Drainage type	Key idea	Example
Antecedent	Older than the mountain	Ganga, Indus, Satluj, Brahmaputra
Consequent	Follows the original slope	Godavari, Krishna, Kaveri
Subsequent	Along weak rock bands	Strike-valley tributaries
Superimposed	Let down onto older rocks	Chambal, Son, Banas

6.2 The Ganga and its Headwaters

- Ganga rises as the **Bhagirathi** at **Gaumukh** (Gangotri glacier); called **Ganga** from **Devprayag** 4× 2002, 2016, 2023, 2024
- **Badrinath** is on the **Alaknanda**; **Kedarnath** on the **Mandakini** 2× 2002, 2013
- **Rudraprayag** = Alaknanda + **Mandakini**; **Karnaprayag** = Alaknanda + **Pindar**; **Tehri** = Bhagirathi + **Bhilangna** 2× 2002, 2024
- Adi Shankaracharya's four mathas: **Jyotirmath, Sringeri, Dwarka, Puri**; **Kedarnath is not** one of them 1× 2002
- In Bangladesh the Ganga is the **Padma**; the Bhagirathi-Hooghly branch flows past Kolkata 1× 2016
- Ganga pollution comes from **sewage and industrial waste** (Kanpur tanneries); "holier = more polluted" is false 1× 2005

PANCH PRAYAG: WHERE THE ALAKNANDA'S TRIBUTARIES JOIN

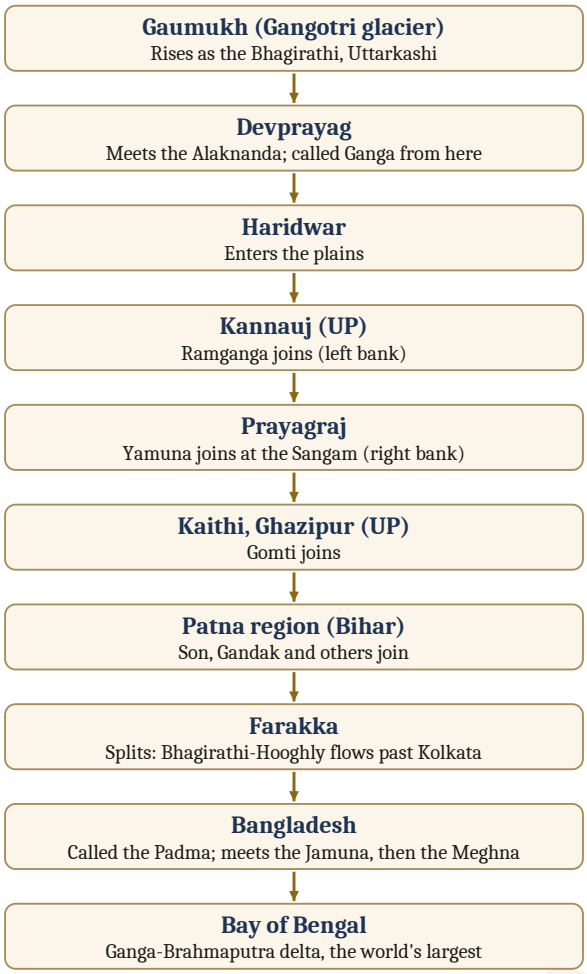


TRAP ALERT

- **Devprayag** (Bhagirathi) and **Rudraprayag** (Mandakini) are the usual swap. Tehri is **Bhagirathi + Bhilangna**, not an Alaknanda confluence.

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COURSE OF THE GANGA, SOURCE TO SEA

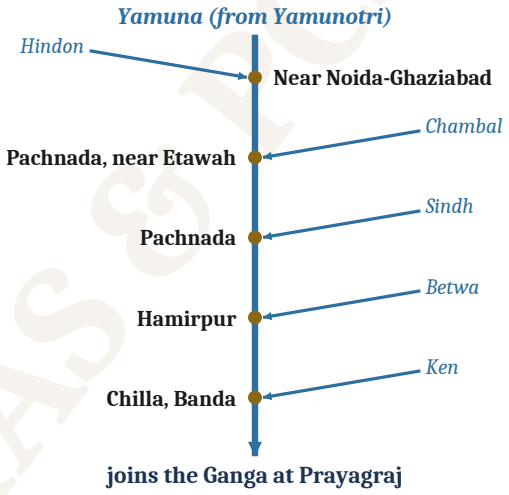


6.3 Tributaries of the Ganga

- **Gomti** is the only Ganga tributary that **rises in the plains**: Phulhar Tal, **Pilibhit**; flows past Lucknow and Jaunpur 1× 2008
- **Yamuna** is **perennial**; it is "dead" below Delhi because water is diverted and sewage added (A true, R false) 1× 2015
- **Ramganga** joins the **Ganga** (at Kannauj), **not** the Yamuna 1× 2010
- Yamuna's right-bank tributaries, **west to east**: **Chambal, Sindh, Betwa, Ken** 1× 2020
- **Son** has the largest catchment among Son, Gandak, Ramganga and Mahananda (about 70,000 sq km) 1× 2017
- **Gandak** does **not** flow through UP; it flows through Bihar 1× 2021
- **Jonk** belongs to the **Mahanadi** basin, not the Ganga 1× 2021
- **Kosi**, the "Sorrow of Bihar", **shifts its course** the most 1× 2013
- **Mahatma Gandhi Setu** (Patna-Hajipur) is in **Bihar** 1× 1993
- **Moribund delta** = western part of the **Bengal (Ganga-Brahmaputra) delta** 1× 2019
- Tributary match: **Ganga-Son, Godavari-Manjira, Krishna-Bhima, Yamuna-Ken** 1× 2003

Tributary	Bank	Joins at
Ramganga	Left	Kannauj (UP)
Gomti	Left	Kaithi, Ghazipur (UP)
Ghaghara	Left	Near Chhapra (Bihar)
Gandak	Left	Sonepur-Hajipur (Bihar)
Kosi	Left	Kursela (Bihar)
Yamuna	Right	Prayagraj (Sangam)
Son	Right	Near Patna

YAMUNA'S TRIBUTARIES, WEST TO EAST (ASKED 2020)



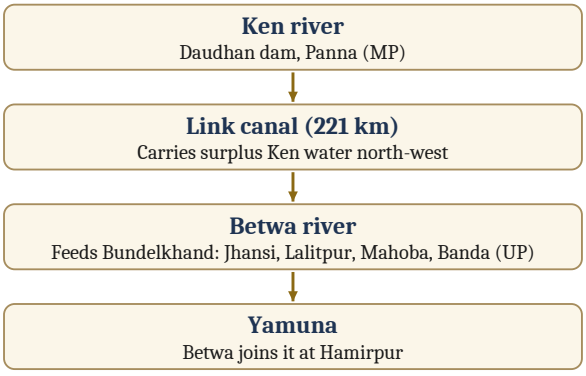
UP LINK

Gomti (Pilibhit to Lucknow to Jaunpur to Kaithi), the Yamuna's Bundelkhand tributaries (Chambal near Etawah, Betwa at **Hamirpur**, Ken at Chilla in **Banda**) and the Kanpur-Prayagraj-Varanasi stretch of the Ganga are UP's most-asked river facts.

6.4 Ken-Betwa and Bundelkhand

- **Ken-Betwa Link**: **UP and MP**, MoU **2005**; India's **first** river-interlinking project; foundation stone **25 Dec 2024** 1× 2006
- **Betwa** joins the **Yamuna** (at Hamirpur) 1× 2014
- **Rajghat dam** (Lalitpur, joint UP-MP project) is on the **Betwa** 2× 2006, 2015

KEN-BETWA LINK: DIRECTION OF WATER



QUANTUM IAS & PCS

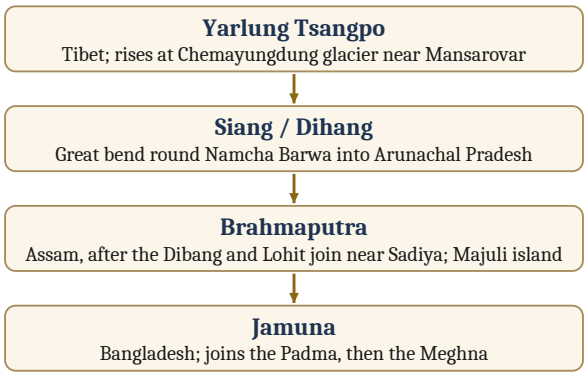
TRAP ALERT

- Water goes **from the Ken to the Betwa** (Daudhan dam on the Ken). Both are tributaries of the **Yamuna**, not the Ganga.

6.5 The Brahmaputra System

- **Indus and Brahmaputra** rise at almost the same point, near **Mansarovar-Kailash** 1× 2006
- Brahmaputra basin: **Tibet, India, Bangladesh** (and Bhutan) 1× 1990
- **Teesta** was once a Ganga tributary; after the **1787 flood** it joined the **Brahmaputra**: **river capture** (A and R true, R explains A) 1× 2021
- **Sankosh** forms the **Assam-West Bengal** boundary 1× 2007

ONE RIVER, FOUR NAMES: THE BRAHMAPUTRA



6.6 The Indus System

- **Satluj** rises at **Mansarovar-Rakas Tal** and enters India at **Shipki La** 1× 2023
- **Satluj does not rise in India**; Beas, Ravi and Chenab do 2× 2010, 2013
- **Kishanganga** (Neelum) is a tributary of the **Jhelum** 1× 2013
- **Doodh Ganga** flows in **Jammu and Kashmir** (also a stream of that name in Uttarakhand) 1× 2015
- World's **highest railway bridge**: on the **Chenab** (Reasi, 359 m; opened June 2025) 1× 2006

River	Source
Jhelum	Verinag spring
Chenab	Chandra + Bhaga at Tandi
Ravi	Near Rohtang
Beas	Beas Kund
Satluj	Rakas Tal (Tibet)

LAST-MINUTE 10

1. Perennial Himalayan rivers = **snow-melt**
2. Ganga = **antecedent**; Chambal = **superimposed**
3. **Satluj**: Rakas Tal, Shipki La, cuts all ranges, trans-Himalayan
4. **Devprayag** = Alaknanda + Bhagirathi
5. **Badrinath** on Alaknanda; **Kedarnath** on Mandakini
6. **Gomti** = only plains-born tributary (Pilibhit)
7. Yamuna tributaries west to east: **Chambal, Sindh, Betwa, Ken**
8. **Ken to Betwa** link, UP-MP; Rajghat dam on Betwa
9. Teesta = **river capture** (1787); Kosi = Sorrow of Bihar
10. Kishanganga to **Jhelum**; Chenab bridge = world's highest

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Chapter 7

Drainage II: Peninsular Rivers and River Cities

Built from 39 UPPSC questions (1992–2025). Focus: rift valleys and estuaries, river sources, Godavari, cities on rivers.

ASKED MOST · REVISE FIRST

- **Narmada and Tapi** flow west through **rift valleys** and form **estuaries**, not deltas **8×** 1999, 2002, 2005, 2006, 2010, 2012, 2013, 2017
- Western Ghat rivers form **no deltas**: short, fast courses over hard rock **3×** 2010, 2013, 2014
- **Godavari** = longest Peninsular river with the largest basin **3×** 2009, 2013, 2014
- **Bhubaneswar is not on the Mahanadi**; **Hyderabad is on the Musi**, not the Krishna **4×** 1996, 1999, 2010, 2016

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

7.1 West-flowing Rivers and Rift Valleys

- Western Ghat rivers make **no delta**: short, swift, hard rock (A and R true, R explains A) **3×** 2010, 2013, 2014
- Narmada and Tapi reach the **Arabian Sea** because they flow in **rift valleys** **2×** 1999, 2002
- The Narmada forms **no delta** (A false, R true) **1×** 2005
- River in a **fault trough**: **Narmada** {2010}; rift valley: **Tapi** **2×** 2013, 2017
- **Narmada** flows **between two ranges** (Vindhya and Satpura) **1×** 2012
- **Tapti** forms an **estuary** **1×** 2006
- **Damodar** is famous for **fault-valley (rift) drainage** on the east side **1×** 2019
- Of Periyar, Pennar and Palar, only the **Periyar** falls into the Arabian Sea **1×** 2025
- **Luni**: fresh in the upper course, **saline** lower down **1×** 2019

WHICH WAY DO THEY FLOW?

West to the Arabian Sea
Narmada, Tapi, Mahi, Sabarmati, Luni, Periyar

↓

Form estuaries
Narmada, Tapi (rift valleys)

↓

East to the Bay of Bengal
Mahanadi, Godavari, Krishna, Kaveri, Pennar

↓

Form deltas
Mahanadi, Godavari, Krishna, Kaveri

7.2 Sources of the Rivers

- **Amarkantak**: source of the **Narmada** (and Son) **1×** 2008
- **Son**'s actual source: **Son Bachharwar**, Bilaspur district **1×** 2007
- **Tapti** does **not** rise in the Western Ghats (it rises at **Multai**, Satpura) {2018}; it falls into the **Arabian Sea** **1×** 1992
- Sources: **Yamuna-Yamunotri, Krishna-Mahabaleshwar, Godavari-Nashik, Mahanadi-Sihawa** **1×** 2023
- **Kaveri** rises in the **Brahmagiri** hills (Talakaveri) **2×** 2011, 2013

River	Source
Narmada, Son	Amarkantak (MP)
Tapi	Multai, Betul (MP)
Mahanadi	Sihawa, Dhamtari (Chhattisgarh)
Godavari	Trimbakeshwar, Nashik
Krishna	Mahabaleshwar
Kaveri	Talakaveri, Brahmagiri
Luni	Near Ajmer (Aravalli)

7.3 Basins and Tributaries

- Longest Peninsular river: **Godavari** {2013, 2014}; largest catchment: **Godavari** **1×** 2009
- Largest basin among the options: **Mahanadi** **2×** 2021, 2022
- "Ganga of the South": **Kaveri** in the UPPSC key (NCERT calls the Godavari Dakshin Ganga) **1×** 2015
- **Pranhita** joins the **Godavari**, not the Mahanadi **1×** 2023
- **Tel** is a Mahanadi tributary, **not** the Krishna's **1×** 2010
- **Hagari** is a tributary of the **Tungabhadra** **1×** 2009
- South Indian rivers are mainly **dendritic** **1×** 2014
- **Mahanadi** forms its delta in **Odisha** **1×** 2014
- **Krishna dispute**: Maharashtra, Karnataka, AP (and Telangana); no option complete (*) **1×** 2013
- Water-deficient basins: options 1 and 2 (per the source key) **1×** 2002

River	Main tributaries
Mahanadi	Seonath, Hasdeo, Ib, **Tel**, Jonk
Godavari	Manjra, Penganga, Wardha, Wainganga, **Pranhita**, Indravati
Krishna	Koyna, Bhima, Tungabhadra, Musi
Tungabhadra	Tunga, Bhadra, **Hagari**
Kaveri	Hemavati, Kabini, Bhavani, Amaravati

7.4 The Damodar and Others

- **Indravati** is **not** a Damodar tributary (it joins the Godavari) **1×** 2019
- Damodar joins the **Hooghly** **2×** 2006, 2008

- **Eden Canal** is taken from the **Damodar** 1× 2015
- Most polluted among the options: **Damodar** ("Sorrow of Bengal", coal belt) 1× 2013
- Mumbai's **Mithi** river rises from **Vihar Lake** 1× 2014

7.5 Cities on River Banks

- **Leh** is on the **right bank of the Indus** 1× 2013
- **Bhubaneswar-Mahanadi** is wrong (Cuttack is on the Mahanadi) 2× 1999, 2010
- **Hyderabad-Krishna** is wrong; Hyderabad is on the **Musi** 2× 1996, 2016
- **Cuttack-Mahanadi, Ludhiana-Satluj, Nashik-Godavari, Ujjain-Kshipra** 1× 1992
- **Ujjain** is on the **Kshipra** 1× 2008
- **Govindghat** is **not** on the Mandakini (it is on the Alaknanda) 1× 2013

City	River
Leh	Indus
Srinagar	Jhelum
Ludhiana	Satluj
Jabalpur	Narmada
Surat	Tapi
Ahmedabad	Sabarmati
Ujjain	Kshipra
Nashik, Rajahmundry	Godavari
Vijayawada	Krishna
Hyderabad	Musi
Cuttack, Sambalpur	Mahanadi
Tiruchirappalli	Kaveri
Kota	Chambal

UP LINK

UP river cities: **Lucknow, Jaunpur-Gomti**; **Kanpur, Varanasi-Ganga**; **Agra, Mathura-Yamuna**; **Ayodhya-Saryu**; **Gorakhpur-Rapti**; **Bareilly, Moradabad-Ramganga**. **Amarkantak** (source of the Son) feeds Sonbhadra's river.

TRAP ALERT

- **Bhubaneswar** is not on any big river; **Cuttack** is the Mahanadi city.
- **Tapi** rises in the **Satpura** (Multai), not the Western Ghats.

LAST-MINUTE 10

1. Narmada, Tapi: **rift valley**, west, **estuary**
2. Western Ghat rivers: **no delta**
3. **Damodar**: rift on the east; joins the Hooghly; Eden Canal
4. **Amarkantak** = Narmada, Son; **Sihawa** = Mahanadi
5. **Mahabaleshwar** = Krishna; **Nashik** = Godavari; **Brahmagiri** = Kaveri
6. **Godavari** = longest Peninsular river
7. Pranhita to Godavari; Tel to Mahanadi; Hagari to Tungabhadra
8. Hyderabad = **Musi**; Ujjain = **Kshipra**; Leh = **Indus**
9. **Luni**: fresh above, saline below
10. **Periyar** = the Arabian-Sea river among Periyar, Pennar, Palar

Chapter 8

Waterfalls, Lakes and Hot Springs

Built from 25 UPPSC questions (1997–2022). Focus: waterfall-river matches, Chilika, lake-state matches, Wular.

ASKED MOST · REVISE FIRST

- **Waterfall-river/state matching**: **Jog-Sharavati, Gokak-Ghataprabha, Dudhsagar-Mandovi (Goa), Duduma-Machkund, Chitrakote-Indravati** 6× 2005, 2014, 2016, 2017, 2021, 2022
- **Lake-state matching**: **Loktak-Manipur, Kolleru-AP, Pulicat-AP/TN, Ashtamudi-Kerala, Roopkund-Uttarakhand, Surajkund and Badkhal-Haryana** 5× 2005, 2010, 2012, 2015, 2020
- **Chilika** (Odisha, **Northern Circars** coast): largest **brackish lagoon** on the east coast 4× 2008, 2016, 2021, 2021
- Highest waterfall (UPPSC key): **Kunchikal**, **455 m** (Karnataka) 2× 2015, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 Waterfalls

- Highest waterfall: **Kunchikal** {2018}; height **455 m** 1× 2015
- **Jog-Krishna** is wrong; Jog is on the **Sharavati** 1× 2005
- **Budha Ghagh-Kanchi** is wrong; Budha Ghagh is on the **Burha**; **Dassam** is on the Kanchi 1× 2021
- Match: **Dudhsagar-Mandovi, Duduma-Machkund, Gokak-Ghataprabha, Jog-Sharavati** 1× 2017
- Match: **Dudhsagar-Goa, Barkana-Karnataka, Khandadhar-Odisha, Palani-Himachal** 1× 2022
- Waterfall in **Goa**: **Dudhsagar** 1× 2014
- **Chitrakote** (the "Niagara of India") is on the **Indravati** {2016}, in **Bastar** 1× 2017

Waterfall	River	State
Kunchikal	Varahi	Karnataka
Jog	Sharavati	Karnataka
Gokak	Ghataprabha	Karnataka
Barkana	Sita	Karnataka
Sivasamudram	Kaveri	Karnataka
Dudhsagar	Mandovi	Goa
Duduma	Machkund	Odisha-AP
Khandadhar	Korapani	Odisha
Chitrakote	Indravati	Chhattisgarh
Dhuandhar	Narmada	MP
Hundru	Subarnarekha	Jharkhand
Dassam	Kanchi	Jharkhand
Palani	Local stream	Himachal

TRAP ALERT

- **Palani waterfall** is in **Himachal** (Kullu), not the Palani hills of Tamil Nadu.
- **Nohkalikai** (Meghalaya) is the tallest single plunge in many sources; pick Kunchikal when it is offered.

8.2 Coastal Lagoons and Southern Lakes

- **Vembanad** lake is in **Kerala** 1× 1997
- Largest saline (brackish) lake: **Chilika**; largest **inland** salt lake: **Sambhar** 1× 2021
- **Chilika** lies on the **Northern Circars** coast (Odisha) 2× 2008, 2016
- Largest lagoon on the east coast: **Chilika** 1× 2021
- Lake shared by two states: **Pulicat** (Andhra Pradesh and Tamil Nadu) 1× 2008
- **Loktak-Meghalaya** is wrong; Loktak is in **Manipur** 1× 2005
- **Kolleru-Andhra Pradesh** is correct 2× 2010, 2012
- Match: **Ashtamudi-Kerala, Pulicat-Tamil Nadu, Roopkund-Uttarakhand, Surajkund-Haryana** 1× 2015
- Match: **Sala-Arunachal, Badkhal-Haryana, Loktak-Manipur, Kaliveli-Tamil Nadu** 1× 2020

8.3 Himalayan and Inland Lakes

- Largest **freshwater** lake: **Wular** (J&K; Jhelum) 1× 2021
- **Anchar** lake is in **Jammu and Kashmir** (Srinagar) 1× 2015
- "Mysterious lake" (skeletons): **Roopkund** (Uttarakhand) 1× 2015
- **Ghepan** lake is in **Himachal Pradesh** (Lahaul) 1× 2016
- **Chapanala** lake is in **Assam** 1× 2014
- **Fulhar** lake is in **Uttar Pradesh** (Pilibhit; source of the Gomti) 1× 2009

Lake	State / UT
Wular, Dal, Anchar	Jammu and Kashmir
Pangong Tso, Tso Moriri	Ladakh
Roopkund, Nainital	Uttarakhand
Ghepan, Chandratal	Himachal Pradesh
Sala (Sela)	Arunachal Pradesh
Chapanala	Assam
Loktak	Manipur
Sambhar, Pushkar	Rajasthan
Badkhal, Surajkund	Haryana

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Lake	State / UT
Lonar	Maharashtra
Kolleru, Pulicat	Andhra Pradesh
Kaliveli	Tamil Nadu
Vembanad, Ashtamudi	Kerala
Chilika	Odisha
Fulhar	Uttar Pradesh

8.4 Hot Springs

- **Gujarat-Narayani** is wrong; **Narayani** is in **Rajasthan**; Gujarat has Tulsishyam and Unai 1× 2021

Hot spring	State
Manikaran, Tattapani	Himachal Pradesh
Puga, Panamik	Ladakh
Taptapani, Atri	Odisha
Tulsishyam, Unai	Gujarat
Bakreshwar	West Bengal
Rajgir	Bihar
Sohna	Haryana
Narayani	Rajasthan

UP LINK

Fulhar (Phulhar) Tal in Pilibhit is UP's only lake asked so far: it is the source of the **Gomti**. UP has Ramsar sites such as **Nawabganj, Sandi, Samaspur, Parvati Arga, Saman, Sarsai Nawar, Sur Sarovar (Keetham)** and **Haiderpur**.

LAST-MINUTE 10

1. **Kunchikal** = highest waterfall (455 m)
2. **Jog** = Sharavati; **Gokak** = Ghataprabha
3. **Dudhsagar** = Mandovi, Goa
4. **Chitrakote** = Indravati, Bastar
5. **Chilika** = largest brackish lagoon, Northern Circars
6. **Sambhar** = largest inland salt lake
7. **Wular** = largest freshwater lake
8. **Loktak** = Manipur; **Pulicat** = AP-TN
9. **Roopkund** = mysterious lake; **Fulhar** = UP (Gomti)
10. **Narayani** hot spring = Rajasthan

Quantum IAS & PCS

Chapter 9

Climate and the Monsoon

Built from 25 UPPSC questions (1993–2025). Focus: monsoon branches, winter rain, rainfall extremes, local storms.

ASKED MOST · REVISE FIRST

- **Winter rain**: north-west India from **western disturbances**; **Tamil Nadu** from the **north-east monsoon** 7× 1993, 2008, 2009, 2012, 2019, 2021, one-liner
- **Leh** gets the **least rainfall** among the options 3× 2008, 2009, 2010
- India is a **monsoon** land; the **Himalaya** gives it climatic distinctiveness 2× 1996, 1998
- **Tamil Nadu** coast stays **dry** in the south-west monsoon (parallel to the Bay branch, rain shadow of the Arabian branch) 1× 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

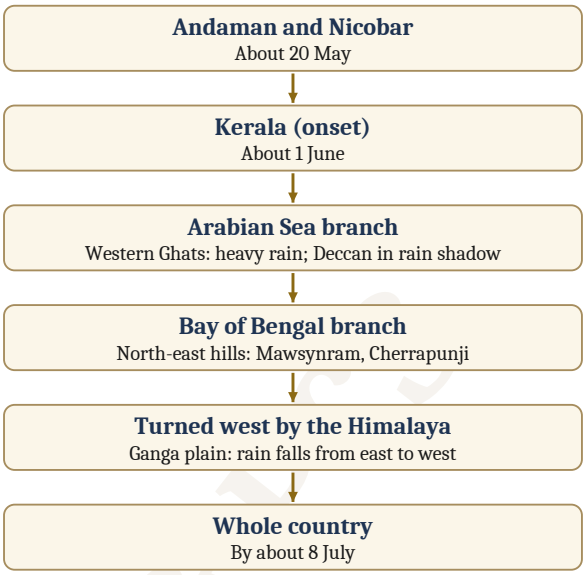
9.1 Monsoon Land and the Himalaya

- India is **monsoonal** because of the **Himalaya** (A and R true, R explains A) 2× 1996, 1998
- Himalayan vegetation varies because **climate changes with altitude** 1× 2025
- **Southern** Himalayan slopes are **wetter and greener**; the claim that northern slopes are greener is false 1× 2023
- **Agra** (16°C) is warmer than **Darjeeling** (4°C) in January because temperature **falls with altitude** 1× 2022

9.2 Arrival and Branches of the Monsoon

- Monsoon reaches **Kerala** first on the mainland (Andaman and Nicobar earliest overall) 1× 2006
- The Indian monsoon comes from the **south-west winds** 1× 2016
- **Tamil Nadu** coast is **dry** in June-September (A and R true, R explains A) 1× 2023

HOW THE SOUTH-WEST MONSOON SPREADS



9.3 Rainfall

- Highest average annual rainfall among the options: **Sikkim** 1× 2016
- Humid places among the options: **Kochi and Tezpur** 1× 2010
- ■ Least rainfall: **Leh** 3× 2008, 2009, 2010
- "200 mm average" had no correct group (probably meant 200 cm) (*) 1× 1994
- India is a "thirsty land" despite heavy rain for **all the reasons given** (seasonal, uneven, run-off) 1× 1994
- Cost-effective water strategy: **rainwater harvesting** 1× 2011
- Inland waterways are weak because rain falls in **only four months** (R explains A) 1× 2008

Annual rainfall	Regions
Over 200 cm	West coast, north-east, Andaman and Nicobar
100-200 cm	East India, middle Ganga plain
50-100 cm	West UP, Punjab, Haryana, inner Deccan
Below 50 cm	West Rajasthan, Kutch, Ladakh

9.4 Winter Weather

- Winter is **dry** for most of India; NE trade winds (source key: R true but not the explanation) 1× 2023
- Most **north-east monsoon** rain: **Tamil Nadu** {2009}; Tamil Nadu's winter rain comes from it 1× 2008
- Winter rain in the **northern plains**: **western disturbances** 2× 1993, 2021
- Western-disturbance rain **decreases from west to east** 1× 2019
- Area with winter rain: the **north-west** 1× 2012
- Northern plains get winter rain, but **not because of the NE monsoon** (R not the explanation) 1× one-liner

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TRAP ALERT

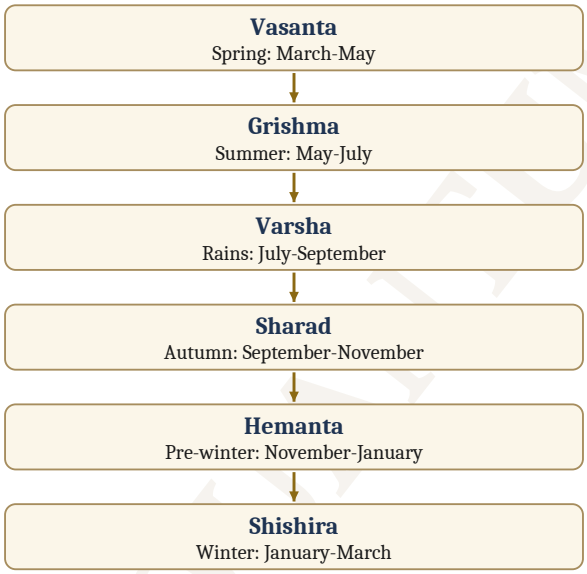
- **North-west** winter rain = **western disturbance**. **South-east** (Tamil Nadu) winter rain = **north-east (retreating) monsoon**. Never mix them.

9.5 Local Storms and Seasons

- Match: **Mango shower-Kerala, Blossom shower-coastal Karnataka, Kal Baisakhi-West Bengal, Loo-Punjab to Bihar** 1× 2024
- Local storm of **Bengal and Assam**, good for tea and rice: **Norwesters (Kal Baisakhi)** 1× 2023
- Hindu seasons in order: **Vasanta (spring), Grishma (summer), Varsha (rain), Sharad (autumn)** 1× 2021

Local wind	Region	Effect
Loo	Punjab to Bihar	Hot, dry, May-June
Kal Baisakhi	West Bengal, Assam	Thunder squalls; tea, jute, rice
Mango shower	Kerala, coastal Karnataka	Ripens mangoes
Blossom shower	Kerala, Karnataka	Coffee flowers
Western disturbance	North-west	Winter rain and snow

SIX HINDU SEASONS (RITU)



UP LINK

UP's rainfall falls from **east to west**: eastern UP gets about 100-150 cm, western UP about 60-80 cm, because the Bay of Bengal branch weakens as it moves up the Ganga plain. The **Loo** blows across UP in May-June; western UP gets **western-disturbance** rain in winter.

LAST-MINUTE 10

1. **Himalaya** makes India a monsoon land
2. Monsoon reaches **Kerala** first (Andamans earliest)
3. Monsoon = **south-west** winds
4. **Tamil Nadu** dry in the SW monsoon; wet from the **NE monsoon**
5. North-west winter rain = **western disturbance**, decreasing west to east
6. **Leh** = least rain; **Mawsynram** = most
7. **Loo** = Punjab to Bihar; **Kal Baisakhi** = Bengal-Assam
8. **Mango shower** = Kerala; **blossom shower** = coffee
9. Temperature falls with **altitude** (Agra vs Darjeeling)
10. Seasons: Vasanta, Grishma, Varsha, Sharad, Hemanta, Shishira

Chapter 10

Natural Disasters

Built from 15 UPPSC questions (2000–2021). Focus: tsunami and cyclones, seismic zones, flood-prone areas, disaster bodies.

ASKED MOST · REVISE FIRST

- **East coast** has more cyclones than the west, but **not** because of the NE trade winds (R not the explanation) 1× 2003
- **Bihar** is the most flood-prone state; in UP the **eastern** region floods most 2× 2000, 2012
- India has **4 seismic zones** (II to V); zone **V** is the most dangerous 3× 2006, 2013, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Tsunamis and Cyclones

- Worst hit by the 2004 tsunami: **Coromandel coast** (Tamil Nadu) 1× 2014
- Tsunami Warning Centre: **Hyderabad** (INCOIS) 1× 2012
- **IMD** headquarters: **New Delhi** (est. 1875) 1× 2012
- **East coast** is more cyclone-prone (A true; R true but not the reason) 1× 2003
- Cyclone **Tauktae** formed in the **Arabian Sea** 1× 2021

Cyclone	Year	Sea	Landfall
Fani	2019	Bay of Bengal	Odisha
Amphan	2020	Bay of Bengal	West Bengal
Tauktae	2021	Arabian Sea	Gujarat
Yaas	2021	Bay of Bengal	Odisha
Biparjoy	2023	Arabian Sea	Kutch
Michaung	2023	Bay of Bengal	Andhra Pradesh
Remal	2024	Bay of Bengal	West Bengal
Fengal	2024	Bay of Bengal	Puducherry

10.2 Earthquakes and Seismic Zones

- **Karnataka plateau** is **not** a high-seismic zone 1× 2006
- India has **4 seismic zones** 1× 2013
- City-zone pairs: Bhuj V, Srinagar V, Chennai III, Hyderabad II (three pairs wrong, *) 1× 2014
- **Koyna** is becoming quake-prone because the dam sits on an **old fault** (reservoir-induced) 1× 2001

SEISMIC ZONES, HIGHEST RISK TO LOWEST

Zone V (very high)
North-east, J&K, Himachal, Uttarakhand, Kutch, north Bihar, Andamans

↓

Zone IV (high)
Delhi, Himalayan foothills, most of the Ganga plain and western UP

↓

Zone III (moderate)
Kerala, Goa, Mumbai, Chennai, central India

↓

Zone II (low)
Most of the Peninsula: Karnataka, Hyderabad

TRAP ALERT

- **Zone I no longer exists**; zone II is the lowest. A Peninsular city like Hyderabad is never zone IV or V.

10.3 Landslides and Floods

- Himalayan **landslides** have increased because of **large-scale mining** (R explains A) 1× 2003
- Most flood-prone state: **Bihar** 1× 2000
- Most flood-hit part of UP: the **eastern** region 1× 2012
- State facing the most natural disasters among the options: **Odisha** 1× 2002
- True: damage is worst in **developing** countries; **Bhopal** gas tragedy was man-made; **mangroves** reduce cyclone damage (not "India is disaster-free") 1× 2018

10.4 Disaster Management Bodies

- India's first disaster management training institute: **Latur** (Maharashtra) 1× 2004

Body	Year	Role
NDMA	2005	Policy; chaired by the PM
NDRF	2006	Rescue battalions
NIDM	1995	Training, New Delhi
INCOIS	2007	Tsunami warning, Hyderabad
IMD	1875	Weather and cyclone warning

UP LINK

Eastern UP (Gorakhpur, Bahraich, Basti, Kushinagar) floods most, from the **Ghaghara, Rapti and Gandak** rising in Nepal. Western UP and **Delhi-NCR** lie in seismic **zone IV**; the Tarai belt near Nepal is high-risk too.

LAST-MINUTE 10

1. 2004 tsunami: **Coromandel coast**
2. Tsunami warning: **Hyderabad (INCOIS)**; IMD: **New Delhi**
3. East coast = more cyclones; **Tauktae** = Arabian Sea
4. **4 zones** (II-V); V most dangerous; Karnataka plateau is low
5. **Koyna** quakes: reservoir on an old fault
6. Himalayan landslides: **mining**
7. **Bihar** most flood-prone; **eastern UP** floods most
8. **Odisha** faces most disasters
9. Mangroves cut cyclone damage
10. First DM training institute: **Latur**

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Chapter 11

Soils

Built from 29 UPPSC questions (1998–2024). Focus: black (regur) soil, laterite, usar and saline soils, erosion.

ASKED MOST · REVISE FIRST

- **Black (regur)** soil: from **Deccan basalt**, best for **cotton**, holds moisture; it is **not** rich in humus **10×** 1998, 1999, 2004, 2010, 2010, 2013, 2014, 2015, 2016, 2018
- **Laterite**: heavy rain and leaching; **poor in lime**; Malabar coast, Maharashtra **4×** 2000, 2001, 2006, 2023
- **Alkaline (usar)** soil: largest area in **UP**; **saline** soil: largest in **Gujarat** **4×** 2007, 2012, 2017, 2019

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

11.1 Black (Regur) Soils

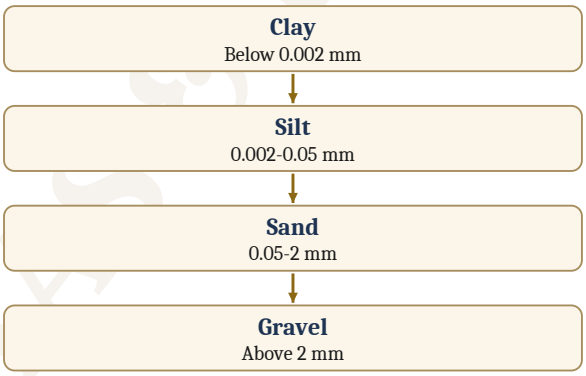
- Formed from weathered **basaltic lava**: **regur** **1×** 2015
- Black cotton soil = **regur** **1×** 2016
- Regur is **black**, but **not** because of humus (A true, R false) **1×** 1999
- Best for cotton: **regur** {2014}; good for cotton, not rich in humus **2×** 2004, 2013
- Lava soils: **Malwa plateau** {1998}; Malwa's main soil: **black** **1×** 2010
- Needs little irrigation because it **retains water**: **black** **1×** 2010
- Wrong statement: "black soil is found in the **Himalaya**" **1×** 2018

11.2 Red, Laterite and Alluvial Soils

- **Red soil**: second-largest group; it surrounds the black soil (both true, R not the explanation) **1×** 2024
- Red colour comes from **diffused iron**, not high iron content; poor in N, P and humus (both true, R not the explanation) **1×** 2023
- Laterite: only statement 1 correct **1×** 2023
- Laterite is **not rich in lime** **1×** 2001
- Laterite found in **Maharashtra** {2006}; mainly on the **Malabar coast** **1×** 2000
- Most productive soil: **alluvial** **1×** 2014
- Match: **Alluvial-UP, Black-Maharashtra, Red-Andhra Pradesh, Desert-Rajasthan** **1×** 2021
- Tea needs **acidic** soil **1×** 2002
- Most deficient micronutrient: **zinc** **1×** 2004
- Himalayan soils are **not** rich in humus overall (A false, R true) **1×** 2004
- Particle under **0.002 mm**: **clay** **1×** 2016

Soil	Rich in	Poor in	Crop
Alluvial	Potash, lime	N, P	Wheat, rice, sugarcane
Black	Lime, iron, potash	N, P, humus	Cotton, soybean
Red	Iron (spread thin)	N, P, humus	Millets, groundnut
Laterite	Iron, aluminium	**Lime**, N, potash	Tea, coffee, cashew
Desert	Salts, phosphate	N, humus	Bajra
Saline/alkaline	Sodium salts	Calcium, N	Salt-tolerant crops

SOIL PARTICLE SIZES, FINEST TO COARSEST



11.3 Saline and Alkaline (Usar) Soils

- Usar soil: **pH above 7**; **paddy can grow** after reclamation; it is reclaimed with **gypsum**, not lime **1×** 2019
- Largest **alkali** area: **Uttar Pradesh** **1×** 2007
- Largest **saline** area: **Gujarat** **2×** 2012, 2017
- Saline soils are unfit for crops because of **dry climate and poor drainage** (R explains A) **1×** 2023

TRAP ALERT

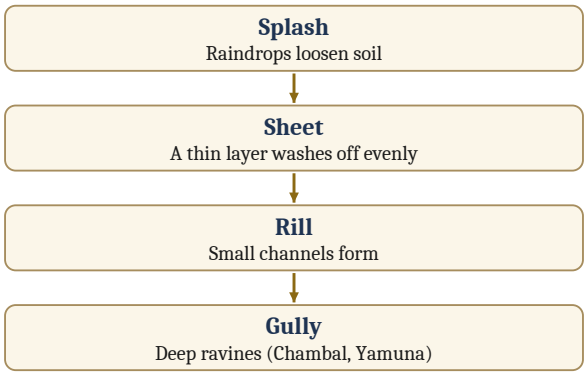
- **Saline** = Gujarat (Rann of Kutch). **Alkaline / usar** = Uttar Pradesh. **Gypsum** reclaims usar, not lime.

11.4 Soil Erosion

- Soil loss, smallest to largest: **splash, sheet, rill, gully** **1×** 2023
- Critical erosion: **Shiwalik foothills** {2004}; worst affected: **Chambal valley** (ravines) **1×** 2004
- Erosion is prevented by **afforestation** **1×** 2014

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EROSION STAGES (LEAST TO MOST SOIL LOSS)



UP LINK

UP has India's largest **usar (alkaline)** area: Raebareli, Unnao, Pratapgarh, Kanpur, Etawah. It is reclaimed with **gypsum**. The **Chambal-Yamuna ravines** (Etawah, Agra, Jalaun) are UP's worst gully-erosion zone. Most of UP is **alluvial** (Bhangar and Khadar); Bundelkhand has red and mixed soils.

LAST-MINUTE 10

1. **Regur** = basalt = cotton = holds water; not humus
2. Lava soil = **Malwa**
3. Red colour = **diffused** iron
4. **Laterite** = no lime; Malabar
5. **Alluvial** = most productive; UP
6. Tea = **acidic** soil; **zinc** most deficient
7. **Usar** = UP, pH > 7, gypsum
8. **Saline** = Gujarat
9. Erosion: splash, sheet, rill, **gully** (Chambal)
10. Clay < 0.002 mm

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Chapter 12

Natural Vegetation and Forests

Built from 12 UPPSC questions (1991–2023). Focus: forest types and their trees, tree line, mangroves.

ASKED MOST · REVISE FIRST

- ■ **Teak and sal** belong to **tropical deciduous** forests (central India, Tarai) 3× 1992, 2007, 2012
- • Forest-region matching: **moist deciduous-Tarai, dry deciduous-middle Ganga plain, alpine-Arunachal, evergreen-Sahyadri** 2× 2001, 2017
- • Mangroves: **Coringa = Godavari delta**; **Sundarbans** = Ganga-Brahmaputra delta 2× 2017, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 Forest Types and Trees

- **Teak and sal**: **tropical dry deciduous** forests 1× 2007
- Teak grows in **MP** (largest), Karnataka, Maharashtra, UP; more than one option fits (*) 1× 2012
- **Deodar-Nilgiri hills** is wrong; deodar grows in the **Himalaya** 1× 2023
- Match: **Teak-central India, Deodar-high Himalaya, Sundari-Sundarbans, Cinchona-Himalayan Tarai** (per key) 1× 1992
- **Cinchona** does **not** grow in **Chhattisgarh** 1× 2010
- Match: **moist deciduous-Tarai, dry deciduous-middle Ganga plain, alpine-Arunachal, evergreen-Sahyadri** 1× 2001
- Match: **tropical-Silent Valley, conifer-Himachal, mangrove-Sundarbans, deciduous-Rajasthan** (per key) 1× 2017
- Plant with **no flowers**: **fern** 1× 1991

FOREST BELTS, WETTEST TO DRIEST

Tropical evergreen
Over 200 cm: Western Ghats, north-east, Andamans

↓

Moist deciduous
100-200 cm: Tarai, sal belt

↓

Dry deciduous
70-100 cm: teak; middle Ganga plain, Deccan

↓

Thorn forest
Below 70 cm: khejri, babool; Rajasthan

12.2 The Himalayan Tree Line

- The tree line is **lower** in the **Western** Himalaya than in the Eastern (cooler and drier in the west) 1× 2005

HIMALAYAN VEGETATION WITH ALTITUDE

Foothills
Tropical moist deciduous: sal

↓

1,000-2,000 m
Subtropical pine (chir), oak

↓

1,500-3,000 m
Temperate: deodar, blue pine, fir, spruce

↓

3,000-3,600 m
Birch, rhododendron, juniper (tree line)

↓

Above 3,600 m
Alpine meadows (bugyals)

12.3 Social Forestry and Mangroves

- Multipurpose tree for **social forestry**: **khejri** 1× 2002
- **Vembanad-Karnataka** is wrong; Vembanad is in **Kerala** 1× 2002
- Mangrove in the **Godavari** delta: **Coringa** 1× 2021

Mangrove	State	Delta / setting
Sundarbans	West Bengal	Ganga-Brahmaputra; largest
Bhitarkanika	Odisha	Brahmani-Baitarani
Coringa	Andhra Pradesh	Godavari
Pichavaram	Tamil Nadu	Kaveri (Coleroon)
Gulf of Kutch	Gujarat	Second largest

TRAP ALERT

- **Coringa** = Godavari; **Pichavaram** = Kaveri; **Bhitarkanika** = Brahmani. **Vembanad** is a Kerala lake.

UP LINK

UP's forests are mostly **tropical dry deciduous** (sal and teak) in the **Tarai** (Dudhwa, Pilibhit) and the **Vindhyan** south (Sonbhadra, Mirzapur). UP has a small forest cover (about 6-7% of its area); **Sonbhadra** has the most forest.

LAST-MINUTE 10

1. **Teak, sal** = tropical deciduous
2. **Deodar** = Himalaya, not Nilgiris
3. Evergreen = **Sahyadri** / Silent Valley
4. Moist deciduous = **Tarai**
5. Tree line **lower in the west**
6. **Khejri** = social forestry
7. **Coringa** = Godavari delta
8. **Sundarbans** = largest mangrove
9. Cinchona not in Chhattisgarh
10. Fern = no flowers

QUANTUM IAS & PCS

Chapter 13

Irrigation and Canals

Built from 17 UPPSC questions (1992–2024). Focus: tube wells in UP, Indira Gandhi Canal, UP canals, canal-river matches.

ASKED MOST · REVISE FIRST

- **Uttar Pradesh** has the largest area under **tube-well** irrigation **4×** 1995, 2006, 2017, one-liner
- **Indira Gandhi Canal** (Rajasthan): from **Harike barrage**; water of the **Satluj, Beas and Ravi**; uses the **Pong** dam on the Beas **5×** 1992, 1999, 2010, 2013, 2024
- **Sharda Sahayak** project aims at more output, multiple cropping and better land management **3×** 2001, 2002, 2008

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

13.1 Sources of Irrigation

- **Ponds (tanks)** are a major source in the Peninsula because its rivers are **seasonal** (R explains A) **1×** 2015
- Highest share of cropped area irrigated (2011) among the options: **Haryana** (Punjab tops overall) **1×** 2015
- Largest tube-well area: **Uttar Pradesh** **3×** 2006, 2017, one-liner
- Tube-well growth was most spectacular in the **Saryupar plain** (eastern UP) **1×** 1995
- Life-saving (protective) irrigation means **PWP irrigation** **1×** 2010

Source	Share	Leading states
Tube wells and wells	About 64%	UP, Punjab, Haryana
Canals	About 24%	Punjab, Haryana, UP, Rajasthan
Tanks	About 3%	Tamil Nadu, Telangana, AP

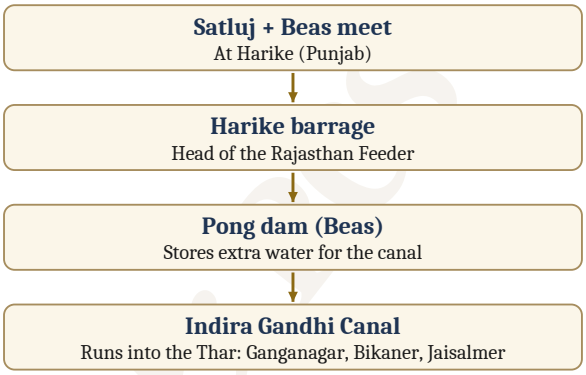
TRAP ALERT

- Highest **percentage** irrigated = **Punjab** (then Haryana). Largest **area** and **tube-well** area = **UP**. Tanks = the **south**.

13.2 Canals and Projects

- **Lower Ganga Canal** starts at **Narora** **1×** 2010
- **Sharda Sahayak** aims: more production, multiple cropping, land reform (all three) **3×** 2001, 2002, 2008
- Match: **Upper Bari Doab-Ravi, Sirhind-Satluj, Gurgaon-Yamuna, Indira Gandhi-Satluj, Ravi, Beas** **1×** 2024
- Indira Gandhi Canal: from **Beas and Satluj** {1992}; source at **Harike barrage** {1999}; water from **Beas, Ravi, Satluj** **1×** 2010
- Project using the **Pong** barrage on the Beas: **Indira Gandhi Canal** **1×** 2013
- **Jawai project-Tamil Nadu** is wrong; Jawai is in **Rajasthan** **1×** 2019
- **Mangalam** irrigation project: **Kerala** **1×** 1998
- **Garland Canal** was proposed by **Dinshaw J. Dastur** **1×** 1999

INDIRA GANDHI CANAL: WHERE ITS WATER COMES FROM



UP canal	Takes off from
Upper Ganga	Ganga at Haridwar (1854)
Lower Ganga	Ganga at **Narora**
Eastern Yamuna	Yamuna at Hathnikund
Agra Canal	Yamuna at Okhla
Sharda Canal	Sharda at Banbasa
Sharda Sahayak	Ghaghara-Sharda link
Saryu Nahar	Ghaghara-Saryu-Rapti (2021)
Betwa Canal	Betwa at Parichha

UP LINK

UP leads India in **tube-well** irrigation and has the **largest irrigated area**. Its big canal systems are the **Upper and Lower Ganga**, **Sharda Sahayak** (largest in UP) and the **Saryu Nahar** (completed 2021, nine eastern districts).

LAST-MINUTE 10

1. **UP** = largest tube-well area; **Saryupar** plain boom
2. **Punjab/Haryana** = highest % irrigated
3. Tanks = Peninsula (seasonal rivers)
4. **Lower Ganga Canal** = Narora; Upper = Haridwar
5. **Indira Gandhi Canal**: Harike; Satluj, Beas, Ravi; Pong dam
6. **Sirhind** = Satluj; **Upper Bari Doab** = Ravi; **Gurgaon** = Yamuna
7. **Jawai** = Rajasthan; **Mangalam** = Kerala
8. **Garland Canal** = Dastur
9. Protective irrigation = **PWP**
10. **Sharda Sahayak** = UP's largest canal

Chapter 14

Multipurpose River Valley Projects and Hydropower

Built from 52 UPPSC questions (1991–2025). Focus: dam-river and dam-state matches, Nagarjuna Sagar, Sardar Sarovar, Indus projects, UP projects.

ASKED MOST · REVISE FIRST

- **Nagarjuna Sagar** is on the **Krishna** (not the Godavari or Kaveri), in **Andhra Pradesh-Telangana** 7× 1991, 1993, 2004, 2004, 2008, 2011, 2012
- **Dam-river matches**: **Hirakud-Mahanadi, Matatila-Betwa, Pochampad-Godavari, Idukki-Periyar, Gandhi Sagar-Chambal, Ukai and Kakrapar-Tapi, Rihand-Rihand** 7× 1995, 2002, 2003, 2014, 2016, 2017, 2019
- **Tulbul** = **Jhelum**; **Baglihar** = **Chenab** (both raised by Pakistan) 6× 2000, 2007, 2008, 2009, 2016, 2025
- **Sardar Sarovar** (Narmada): benefits **Gujarat, MP, Maharashtra, Rajasthan**; MP gets the largest share of power 4× 1992, 1993, 1999, 2004
- First hydropower station: **Sidrapong, Darjeeling** (1897); **Sivasamudram** (Karnataka, 1902) is the oldest big one 6× 1998, 2008, 2008, 2009, 2011, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Hydropower

- Huge hydro potential in the Himalaya; the Nepal border does **not** stop eastern UP and Bihar (only statement 1 true) 1× 2025
- Oldest hydropower station among the options: **Sivasamudram** 1× 2008
- Oldest power plant: **Sidrapong** {1998}; first station at **Darjeeling** 1× 2013
- **Sivasamudram** is in **Karnataka** (not Kerala) 3× 2008, 2009, 2011
- Bhutan plant opened by PM Modi (2019): **Mangdechhu** 1× 2019

14.2 Bhakra and the Indus Projects

- **Bhakra-Nangal**: joint project of **Punjab, Haryana and Rajasthan** 1× 1994
- **Kol Dam** is on the **Satluj** 1× 2017
- **Siul (Baira Siul)** project: **Himachal Pradesh** 1× 2010
- **Baglihar** is on the **Chenab** 2× 2008, 2009
- **Tulbul** is on the **Jhelum** (Wular barrage) 3× 2000, 2007, 2016
- Wrong pairs: **Pandoh-Ravi** (it is on the Beas) and **Srisailam-Tungabhadra** (it is on the Krishna) 1× 2025

Project	River
Bhakra, Nangal, Kol, Nathpa Jhakri	Satluj
Pong, Pandoh	Beas
Ranjit Sagar, Baira Siul	Ravi (system)
Baglihar, Dulhasti, Salal	Chenab
Tulbul, Kishanganga	Jhelum

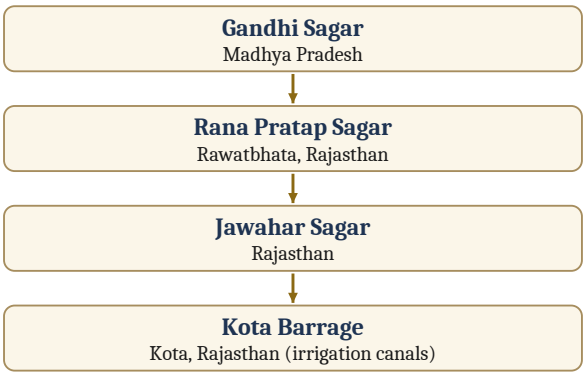
14.3 The Narmada Projects

- **Sardar Sarovar** beneficiaries: **Gujarat, Maharashtra, MP, Rajasthan** 2× 1999, 2004
- Sardar Sarovar is on the **Narmada** 1× 1992
- Most benefit (by water and power share): **Madhya Pradesh** 1× 1993
- **Harsud** town was submerged by the **Indira Sagar** reservoir 1× 2003
- **Tawa project-Krishna** is wrong; the Tawa is a **Narmada** tributary 1× 2015

14.4 Chambal, Damodar and Others

- **Gobind Sagar** (Bhakra) and **G.B. Pant Sagar** (Rihand) are **not** Chambal projects 1× 2011
- **Govind Ballabh Pant Sagar** is in **Uttar Pradesh** (Sonbhadra) 1× 2013
- Benefits shared by more than one state: **Chambal Valley** (MP and Rajasthan) 2× 2007, 2009
- **Kaveri dispute**: **Tamil Nadu, Karnataka, Kerala, Puducherry** 1× 2010
- **Damodar** was Bengal's "River of Sorrow": fast upstream, sluggish downstream (R explains A) 1× 2010
- **Tilaiya-Konar** is wrong; Tilaiya is on the **Barakar** 1× 2013
- Match: **Tilaiya-Barakar, Panchet-Damodar, Rana Pratap Sagar-Chambal, Matatila-Betwa** 1× 2014
- All true: Rihand on a **Son** tributary; Hirakud on the **Mahanadi**; Tungabhadra is **AP-Karnataka**; Maithon on the **Barakar** 1× 2016

CHAMBAL VALLEY PROJECT, UPSTREAM TO DOWNSTREAM



QUANTUM IAS & PCS

TRAP ALERT

- **Rana Pratap Sagar** is in **Rajasthan**, not MP; **Gandhi Sagar** is in MP.
- **Tilaiya and Maithon** = Barakar; **Panchet** = Damodar; **Konar** = Konar.

14.5 Peninsular Projects

- **Nagarjuna Sagar**: **Andhra Pradesh** {1993}; on the **Krishna** {1991, 2004, 2008, 2011}; not Godavari {2012}; not Kaveri 1× 2004
- Match: **Bhakra-Satluj, Hirakud-Mahanadi, Idukki-Periyar, Nagarjuna Sagar-Krishna** 1× 1995
- Match: **Idukki-Periyar, Matatila-Betwa, Nagarjuna Sagar-Krishna, Pochampad-Godavari** 2× 2002, 2003
- Match: **Rihand-Rihand, Nagarjuna Sagar-Krishna, Pochampad-Godavari, Matatila-Betwa** 1× 2016
- Match: **Mettur-Tamil Nadu, Mayurakshi-West Bengal, Nagarjuna Sagar-AP, Hirakud-Odisha** 1× 2008
- Match: **Farakka-WB, Ghataprabha-Karnataka, Hirakud-Odisha, Kakrapar-Gujarat** 1× 2003
- Match: **Dulhasti-Chenab, Gandhi Sagar-Chambal, Ukai-Tapi, Tawa-Tawa** 1× 2017
- Match: **Betwa-Matatila, Mahanadi-Hirakud, Chambal-Gandhi Sagar, Tapi-Kakrapar** 1× 2014
- Match: **Bhadra-Karnataka, Bhavani Sagar-Tamil Nadu, Gandhi Sagar-MP, Rana Pratap Sagar-Rajasthan** 1× 2006
- **Rana Pratap Sagar-MP** is wrong (Rajasthan) 1× 2004
- **Ukai-Tapi, Jayakwadi-Godavari, Khadakwasla-Mutha** 1× 2019
- **Telugu Ganga**: joint project of **TN, AP, Karnataka** {1999}; supplies **Chennai** 1× 1996
- **Babli** project (Godavari dispute): **Maharashtra** 1× 2012
- **Teesta Low Dam III**: **West Bengal** 1× 2003
- **Kalpong** hydro project: **Andaman and Nicobar** 1× 2002
- **Kalpasar** freshwater reservoir: **Gujarat** (Gulf of Khambhat) 1× 2010
- **Malampuzha** dam is on the **Malampuzha** river {2017}; **Cheruthoni** dam on the **Periyar** 1× 2017

Project	River	State
Hirakud	Mahanadi	Odisha
Nagarjuna Sagar, Srisailam	Krishna	AP-Telangana
Pochampad, Jayakwadi, Polavaram	Godavari	Telangana, Maharashtra, AP
Mettur, Krishna Raja Sagar	Kaveri	TN, Karnataka
Idukki, Cheruthoni	Periyar	Kerala
Ukai, Kakrapar	Tapi	Gujarat
Bhadra	Bhadra	Karnataka
Bhavani Sagar	Bhavani	Tamil Nadu
Mayurakshi	Mayurakshi	West Bengal

14.6 UP and Neighbourhood Projects

- **Tehri** dam is on the **Bhagirathi** 1× 2002
- Ramganga project: the **825.8 m length** pair is wrong (about 715 m) 1× 2024
- Match: **Rihand-Son, Rani Laxmibai (Rajghat)-Betwa, Tehri-Bhagirathi, Ramganga-Ramganga** 1× 2010
- **Singrauli** small hydro project: **Uttar Pradesh** 1× 2017

UP LINK

UP projects: **Rihand dam** (G.B. Pant Sagar, Sonbhadra) on the Rihand, a **Son** tributary; **Obra**; **Rajghat** (Rani Laxmibai Sagar) and **Matatila** on the **Betwa** (UP-MP); **Ramganga (Kalagarh)** irrigates western UP; **Tehri** was built when Uttarakhand was part of UP.

LAST-MINUTE 10

1. **Nagarjuna Sagar** = Krishna, AP-Telangana
2. **Hirakud** = Mahanadi; **Pochampad** = Godavari; **Idukki** = Periyar
3. **Matatila, Rajghat** = Betwa; **Rihand** = Son tributary (UP)
4. **Tulbul** = Jhelum; **Baglihar** = Chenab; **Pandoh** = Beas
5. **Bhakra** = Satluj; Punjab, Haryana, Rajasthan
6. **Sardar Sarovar** = Narmada; MP most power; Indira Sagar drowned Harsud
7. **Gandhi Sagar** MP; **Rana Pratap Sagar** Rajasthan
8. **Tilaiya, Maithon** = Barakar; Damodar = Sorrow of Bengal
9. **Telugu Ganga** = water to Chennai
10. **Sidrapong** (Darjeeling) first; **Sivasamudram** Karnataka

Chapter 15

Agriculture and the Green Revolution

Built from 33 UPPSC questions (1994–2024). Focus: colour revolutions, Green Revolution facts, agricultural regions, Pantnagar.

ASKED MOST · REVISE FIRST

- **Yellow Revolution** = **oilseeds** 4× 2002, 2004, 2014, 2017
- **Colour revolutions** (match): Green-food grains, White-milk, Blue-**fish** (not poultry), Grey-fertilisers, Pink-onion 5× 2010, 2010, 2014, 2015, 2016
- **Wheat** gained most from the Green Revolution 3× 1994, 2001, 2015
- India has **20 agro-ecological regions** 3× 2010, 2013, 2016
- First agricultural university: **Pantnagar** (GBPUAT, **1960**) 4× 2010, 2013, 2014, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Regions, Scholars and Institutions

- *The History of Indian Agriculture*: **M.S. Randhawa** 1× 2015
- Agro-ecological regions: **20** 3× 2010, 2013, 2016
- Sengupta and Sdasyuk (1968): **60** micro agricultural regions 1× 2017
- First agricultural university: **Pantnagar, 1960** (true); the "six colleges in 1901-05" claim is treated as false 1× 2023
- Year: **1960** {2013}; name: **GBPUAT, Pantnagar** 1× 2014
- **Zero-till seed-cum-fertiliser drill** was developed at **Pantnagar** 1× 2010

Scheme	By	Number
Agro-ecological regions	NBSS&LUP (ICAR)	**20**
Agro-climatic zones	Planning Commission (1989)	**15**
Micro agricultural regions	Sengupta and Sdasyuk	**60**
NARP zones	ICAR	127

15.2 Structure and Problems

- Indian agriculture is mainly **a means of livelihood** 1× 2004
- True: India has more **cultivated area** than the USA, China, Japan; tropical monsoon allows **year-round cropping** 1× 2024
- Low productivity has **all the given causes** (small holdings, rain dependence, old methods) 1× 2007
- **Cooperative farming** is **not** a cause of low productivity 1× 2003
- Largest average holding among the options: **Punjab** (Nagaland largest overall) 1× 2010
- Dry zone is agrarian; it has second Green Revolution potential (both true, R not the explanation) 1× 2010
- High productivity in the south: **Tamil Nadu coast** 1× 2004
- Richest in replenishable **groundwater**: **Uttar Pradesh** 1× 2006
- Pioneer in **contract farming**: **Punjab** 1× 2005

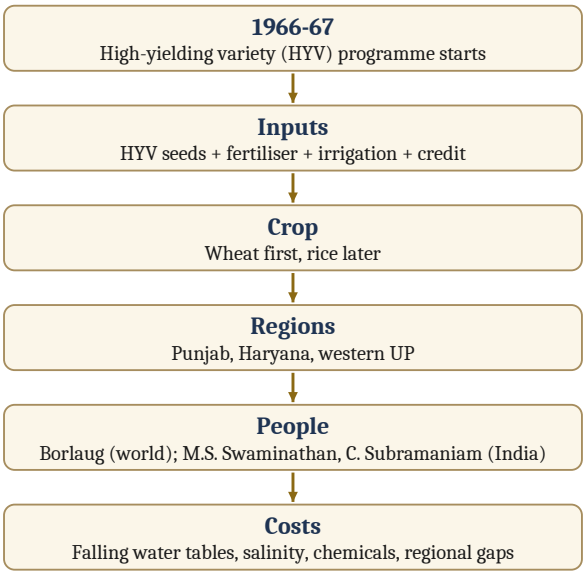
15.3 Fertilisers and Green Manure

- Largest fertiliser users (at that time): **UP and Andhra Pradesh**; per hectare: **Punjab** 1× 2009
- **Balanced fertilisers** serve all the stated purposes 1× 2008
- Green manure with the most nitrogen among the options: **cowpea (lobia)** 2× 2016, 2008
- **Green agriculture** = integrated pest management, nutrient supply and resource management 1× 2016

15.4 The Green Revolution

- New Agricultural Strategy: **1966-67** (key says "seventies", meaning the seventh decade) 1× 2015
- World "Father of the Green Revolution": **Norman Borlaug** 1× 2015
- "**Evergreen Revolution**": **M.S. Swaminathan** 1× 2015
- Green Revolution = **wheat** production 1× 2015
- Biggest gain in output and yield: **wheat** 2× 1994, 2001
- Wheat yield reached about **2,000 kg per hectare** 1× 2003
- It modernised farming; the claim of little social or environmental cost is debatable (key: both true) 1× 2014

GREEN REVOLUTION IN BRIEF



15.5 Colour Revolutions

- Match: **food grains-Green, milk-White, fisheries-Blue, fertilisers-Grey** 1× 2014

- **Pink** Revolution: **onion** (UPPSC key) 1× 2010
- **Blue-poultry** is wrong; Blue = **fish** {2015, 2016}; Blue Revolution = **fisheries** 1× 2010

■ **Yellow** = **oilseeds** 4× 2002, 2004, 2014, 2017

Revolution	Product
Green	Food grains
White	Milk
Blue	Fish
Yellow	Oilseeds
Golden	Fruits, honey
Silver	Eggs, poultry
Pink	Onion, prawns, meat
Red	Meat, tomato
Grey	Fertilisers
Brown	Leather, cocoa
Black	Petroleum
Round	Potato
Sweet	Honey

UP LINK

Pantnagar university (GBPUAT, 1960) was set up in UP (now Uttarakhand). **Western UP** was a Green Revolution core region with Punjab and Haryana. UP is the largest fertiliser consumer and the richest in **replenishable groundwater**. Average holding in UP: about **0.73 ha**.

LAST-MINUTE 10

1. **20** agro-ecological regions; **15** agro-climatic zones; **60** micro regions
2. **Pantnagar 1960** = first agricultural university; zero-till drill
3. **Randhawa** = History of Indian Agriculture
4. Green Revolution: **1966-67**, **wheat**, Punjab-Haryana-west UP
5. **Borlaug** (world), **Swaminathan** (India; Evergreen Revolution)
6. **Yellow** = oilseeds; **Blue** = fish; **Grey** = fertilisers; **Pink** = onion
7. **UP** richest in groundwater; **Punjab** largest holdings, contract farming pioneer
8. Green manure: **cowpea (lobia)**
9. Cooperative farming is not a cause of low yield
10. Tamil Nadu coast = high productivity in the south

QUANTUM IAS & PCS

Chapter 16

Major Crops I: Crop Seasons, Wheat and Rice

Built from 44 UPPSC questions (2001–2024). Focus: kharif and rabi crops, wheat varieties and diseases, rice area, yield and production.

ASKED MOST · REVISE FIRST

- **Wheat varieties**: Kalyan Sona, Sonalika, Raj 3077, Pusa Sindhu Ganga, UP-308 are **wheat** 4× 2005, 2015, 2016, 2016
- **Rice**: largest **area** = **UP**; highest **yield** = **Punjab**; production leader = West Bengal (UP in 2024-25) 4× 2006, 2007, 2015, 2017
- **Rabi** crops: wheat, **mustard**, **gram**; **kharif**: rice, **soybean** 4× 2004, 2011, 2012, 2016
- **Rice varieties**: Jaya, Padma, Krishna, Pusa Sugandha-5, Mahi Sugandha, Barani Deep, Pusa RH-10 5× 2006, 2006, 2013, 2015, 2015
- Wheat diseases: **rust**, **Karnal bunt** 3× 2003, 2004, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 Crop Seasons

- **Mustard** is a **rabi** crop {2016}; not kharif 1× 2012
- **Soybean** is a **kharif** crop 1× 2004
- **Gram** is **not** a kharif crop (it is rabi) 1× 2011
- **Zaid** crops in irrigated areas: **moong and urad** 1× 2011
- Rabi crops lead in Punjab-Haryana, but winter rain there comes from **western disturbances**, not cyclones (A true, R false) 1× 2023
- **Rice** is a **transplanted** crop 1× 2017

Season	Sown	Harvested	Crops
Kharif	June-July	Sept-Oct	Rice, maize, bajra, arhar, cotton, soybean, groundnut
Rabi	Oct-Nov	March-April	Wheat, barley, gram, mustard, pea, potato
Zaid	March	June	Melons, cucumber, moong, urad

16.2 Wheat: States and Output

- Top three wheat states: **UP, Punjab, Haryana** (then; MP now second) 1× 2006
- Largest wheat producer (2015-16): **Uttar Pradesh** 1× 2016
- UP ranks first in **wheat and sugarcane** (and rice since 2024-25) (*) 1× 2008
- UP's biggest cropping pattern is **rice-wheat**; wheat's most critical irrigation is **crown root initiation**, not flowering 1× 2013
- Wheat output **increased** from 2006-07 to 2009-10 1× 2009

16.3 Wheat Varieties and Diseases

- Induced mutation: **Sonora-64** in the key (strictly **Sharbati Sonora**, from Sonora-64) 1× 2015
- Dwarfing gene in wheat: **Norin-10** 1× 2013
- **Kalyan Sona** {2005}, **Raj 3077** {2015}, **Pusa Sindhu Ganga** {2016}, **UP-308** {2016} are **wheat**
- **Macaroni (durum)** wheat suits **unirrigated** conditions 1× 2004
- **Triticale** = **wheat × rye** 1× 2014
- Wheat disease: **rust** {2003, 2004}; **Karnal bunt** 1× 2015
- Arhar for double cropping with wheat: **UPAS-120** 1× 2010

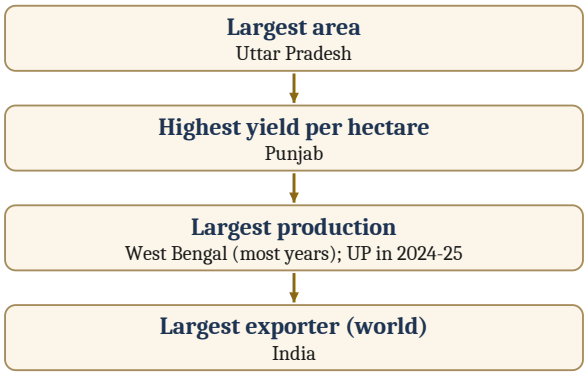
Disease	Crop
Rust, Karnal bunt, loose smut	Wheat
Blast; Khaira (zinc lack)	Rice
Tikka	Groundnut
Red rot	Sugarcane
Wilt	Arhar, gram

16.4 Rice: Area and Production

- Rice originated in **South-East Asia** 1× 2015
- Chief food crop of India: **rice** 1× 2004
- Largest crop area: **rice** 1× 2011
- Rice needs over **100 cm** of rain 1× 2007
- Largest rice **area**: **UP** {2007}; largest producer (then): UP per the key 1× 2006
- Wrong pair: Bihar's share of rice output is **not 15.7%** 1× 2024
- **Aman** paddy is sown in **June-July** (harvested in winter) 1× 2011
- More than half of rice output: the leading states named in the key (*) 2× 2004, 2009
- Four main rice producers (then): **WB, Punjab, UP, AP** (*) 1× 2004
- Ascending order: **Tamil Nadu, Punjab, Andhra Pradesh, West Bengal** 1× 2003
- Highest rice **yield**: **Punjab** 2× 2015, 2017
- Punjab **exports** rice, but it does **not** lead in production (A true, R false) 1× 2001

QUANTUM IAS & PCS

RICE: THREE DIFFERENT "LARGEST" ANSWERS



16.5 Rice Varieties

- **Jaya, Padma, Krishna**: **rice** 1× 2006
- **Pusa Sugandha-5**: aromatic **rice** 1× 2006
- **Mahi Sugandha**: **rice** 1× 2015
- **Barani Deep**: **paddy** 1× 2013
- Hybrid basmati: **Pusa RH-10** 1× 2015
- Seed rate for transplanted basmati: **15-20 kg/ha** 1× 2014
- Largest area under **hybrid rice**: **Uttar Pradesh** 1× 2014

TRAP ALERT

- **Gram and mustard are rabi**; **soybean is kharif**. **Moong and urad** can be kharif or zaid.
- **Norin-10** = wheat dwarfing gene; **Dee-geo-woo-gen** = rice.

UP LINK

UP is India's **largest wheat producer**, has the **largest rice area** and the largest **hybrid rice** area, and became the top **rice producer** in 2024-25. Its main cropping pattern is **rice-wheat**. Wheat variety **UP-308** and arhar **UPAS-120** come from UP-Pantnagar breeding.

LAST-MINUTE 10

1. Rabi: **wheat, mustard, gram**; kharif: **rice, soybean**
2. Zaid: **moong, urad**, melons
3. **Norin-10** = wheat dwarfing gene; **triticale** = wheat × rye
4. Kalyan Sona, Raj 3077, UP-308 = wheat
5. Jaya, Padma, Pusa Sugandha = rice
6. **Rust, Karnal bunt** = wheat
7. Rice: **UP** area, **Punjab** yield, **WB** production (UP 2024-25)
8. Rice needs **100 cm+** rain; originated in SE Asia
9. **Aman** paddy sown June-July
10. UP = largest wheat producer; rice-wheat pattern

Quantum IAS & PCS

Chapter 17

Major Crops II: Cotton, Sugarcane, Oilseeds and Pulses

Built from 44 UPPSC questions (2001–2023). Focus: UP and sugarcane, the southward shift of the sugar industry, soybean, oilseed and pulse varieties.

ASKED MOST · REVISE FIRST

- **UP** leads in sugarcane **area and production**; it is the **"Sugar Bowl"** of India 4× 2008, 2013, 2014, 2015
- Sugar industry shifted **south**: higher cane yield, more sucrose, longer crushing season (R explains A) 2× 2007, 2017
- **MP** = largest **soybean** area and (until 2023-24) production 4× 2001, 2003, 2014, 2015
- **Mustard** varieties: **Varuna, Pusa Bold, Pitambari** 3× 2007, 2013, 2015
- **Arhar** varieties: **Bahar, Malaviya Chamatkar**; Neelam is **not** arhar (it is a mango) 4× 2008, 2008, 2013, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

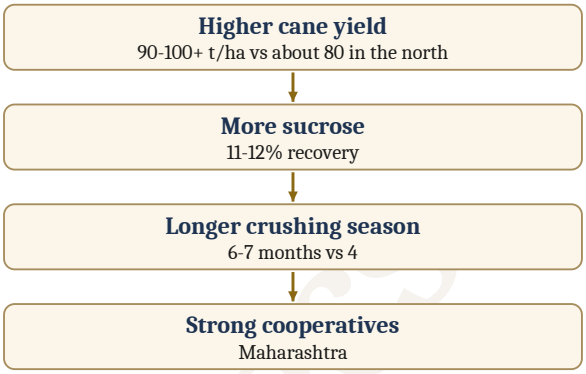
17.1 Cotton

- "**White gold**" of Maharashtra: **cotton** 1× 2002
- Cotton fibre comes from the **seed** (seed hairs) 1× 2009

17.2 Sugarcane and Sugar

- Largest sugarcane area: **UP** {2015}; leading producer: **UP** 1× 2013
- Production order (2013): **UP, Maharashtra, Karnataka, Haryana** 1× 2014
- "**Sugar Bowl**" of India: **UP** 1× 2008
- Crop with the highest **irrigated share** of its area: **sugarcane** 1× 2013
- First sugar mill (1903): **Pratappur** (Deoria, UP) 1× 2013
- Sugarcane breeding: **Coimbatore** (Sugarcane Breeding Institute) 2× 2009, 2010
- **Adsali** cane takes about **18 months** to ripen 1× 2010
- India: over 15% of world sugar, sugar is the 2nd largest agro-industry, India is the **largest consumer**; not the largest producer (Brazil) 1× 2009
- Shift to the south: **higher yield, more sucrose, longer season** (not cheap labour) 1× 2007
- South has more sugar industry because of **higher yield and juice** (R explains A) 1× 2017
- **Sakkar Nagar** (Bodhan) was in AP, now **Telangana** (*) 1× 2008

WHY SUGAR MOVED SOUTH



17.3 Oilseeds

- Leading soybean producer: **Madhya Pradesh** (Maharashtra in 2024-25) 3× 2001, 2003, 2015
- Largest soybean **area**: **MP** 1× 2014
- Largest oilseed producer (2006): **MP**; now **Rajasthan** 1× 2006
- Largest **groundnut** producer: **Gujarat** 1× 2006
- Low area but very high groundnut yield: **West Bengal** 1× 2014
- **Pegging** happens in **groundnut** {2010}; groundnut needs much **gypsum** 1× 2007
- **Kaushal**: groundnut 1× 2002
- **Varuna** {2007}, **Pusa Bold** {2013}, **Pitambari** {2015}: **mustard**
- Highest oil content: **sesame** 1× 2017
- **Kusum** (safflower) = *Carthamus tinctorius* 1× 2016

Oilseed	Oil %	Leading state
Sesame	46-52 (highest)	WB, Gujarat
Groundnut	45-50	Gujarat
Mustard	38-42	Rajasthan
Soybean	18-20	MP, Maharashtra
Safflower	30-35	Maharashtra
Castor	45-50	Gujarat

17.4 Pulses

- India is the largest **producer and consumer** of pulses 1× 2014
- India does **not** usually export **pulses** (it imports them) 1× 2013
- Largest pulse producer: **Madhya Pradesh** 1× 2004
- India lacks pulses but not protein (R explains A, per key) 1× 2009
- **Green gram (moong)** is **not** a rabi pulse 1× 2023
- **Legumes** fix nitrogen from the air 1× 2006

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- Nutrient essential for nodules: **cobalt** 1× 2004
- Poor nitrogen fixer: **rajma** (*) 1× 2009
- NPK ratio for legumes: **1:2:2** 1× 2013
- Pulses complement cereals in intercropping (both true, R not the explanation per key; the real reason is nitrogen fixation) 1× 2023
- **Red gram (arhar)** originated in **India** 1× 2007
- **Malaviya Chamatkar**: **arhar** {2014}; **Bahar**: **arhar** 2× 2008, 2013
- **Arhar-Neelam** is wrong 1× 2008
- Leafless pea: **Aparna** 1× 2013

Kharif pulses	Rabi pulses
Arhar (tur)	Gram (largest pulse)
Moong (green gram)	Masoor (lentil)
Urad (black gram)	Pea
Lobia (cowpea)	Rajma, khesari

UP LINK

UP is the **Sugar Bowl**: top in cane area and output; India's first sugar mill was at **Pratappur** (Deoria, 1903). UP-bred varieties: arhar **Bahar** and **Narendra Arhar** (Ayodhya), **Malaviya Chamatkar** (BHU), pea **Azad** (Kanpur), groundnut **Kaushal, Chandra** (Kanpur).

LAST-MINUTE 10

1. Cotton = "white gold"; fibre from the **seed**
2. **UP** = Sugar Bowl; first mill **Pratappur 1903**
3. Cane breeding at **Coimbatore**; Adsali = **18 months**
4. Sugar moved south: yield, sucrose, longer season
5. **MP** = soybean and pulses; **Gujarat** = groundnut; **Rajasthan** = oilseeds now
6. **Sesame** = highest oil; groundnut = pegging, gypsum
7. **Varuna, Pusa Bold, Pitambari** = mustard
8. **Bahar, Malaviya Chamatkar** = arhar; **Aparna** = leafless pea
9. Moong is **not** rabi; legumes fix nitrogen; **cobalt** for nodules
10. India: largest producer, consumer and importer of pulses

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Chapter 18

Silk, Plantation Crops and Spices

Built from 34 UPPSC questions (1994–2023). Focus: silk types and states, tea, coffee, rubber, spices, commodity boards.

ASKED MOST · REVISE FIRST

- **Karnataka** = largest **coffee** producer (over 70%) and largest coffee area 6× 2002, 2002, 2004, 2006, 2010, one-liner
- **Kerala** = largest **rubber** producer; rubber is never Himachal 4× 2004, 2014, 2017, 2022
- **Karnataka** leads in **mulberry silk**, silk yarn and fabric 4× 2010, 2013, 2017, 2017
- **Kerala** = "**Garden of Spices**"; leads in cardamom and pepper 4× 2010, 2011, 2013, 2018
- **Muga** silk comes only from **Assam** (India); **Eri** = Assam 3× 2008, 2008, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Silk

- Over three-quarters of raw silk came from **Karnataka and AP** in 2006 (no longer true, *) 1× 2006
- Leading **mulberry** silk state: **Karnataka** 1× 2017
- **Eri silk-Assam** is correct 1× 2008
- True: Karnataka = largest raw silk producer; **Jharkhand** = largest **tasar** producer 1× 2010
- **Muga** silk is found only in India, in **Assam** 1× 2008
- Largest silk **fabric** producer: **Karnataka** {2013}; most silk **yarn**: **Karnataka** 1× 2017

Silk	Silkworm food	Leading state
Mulberry (70%)	Mulberry	Karnataka
Tasar	Arjun, sal	Jharkhand
Eri	Castor	Assam
Muga (golden)	Som, soalu	Assam (India only)

18.2 Tea

- Largest tea producer: **Assam** 1× 1994
- Over 200 cm rain on sloping hills: ideal for **tea** 1× 2010
- India was **first** in tea output and use in 2007-08; **China** leads now, India second 1× 2008
- India exports much tea, but home consumption is **not** poor (A true, R false) 1× 2007
- "**Green Gold**" = **tea** 1× 2008
- Tea is a **stimulant** crop 1× 2017

18.3 Coffee and Rubber

- Largest coffee producer: **Karnataka** {2002, 2004, 2006, 2010}; over **70%** of output 1× 2016
- Largest coffee **area**: **Karnataka** 1× 2002
- Largest rubber producer: **Kerala** {2004, 2014}; **Kerala-Rubber** correct 1× 2022
- **Rubber-Himachal** is wrong 1× 2017

18.4 Plantations, Tobacco and Coconut

- Main plantation crops: **tea, rubber, coconut, coffee** 1× 2001
- True: **tea = Assam**, **tobacco = Andhra Pradesh**; coffee is not Tamil Nadu's main crop; jute is not Chhattisgarh's 1× 2001
- Over a third of tobacco: **Andhra Pradesh** (then); **Gujarat** now 1× 2006
- Largest tobacco producer per the 2007 key: **Uttar Pradesh** 1× 2007
- Largest tobacco area (2010): **Andhra Pradesh**; **Gujarat** now 1× 2012
- Largest coconut producer: **Kerala** (then); **Karnataka** since 2023-24 2× 2006, 2014

18.5 Spices, Cashew and Boards

- India leads in **ginger** (and turmeric, chilli); not in cardamom or cloves now (*) 1× 2010
- **Odisha** is **not** a cardamom state 1× 2014
- ■ "**Garden of spices**": **Kerala** 3× 2010, 2011, 2013
- Largest cardamom and pepper producer: **Kerala** 1× 2018
- Biggest spice producer (2012 key): **Madhya Pradesh** 1× 2012
- Cloves grow in Tamil Nadu, Kerala, Karnataka (*) 1× 2007
- Major **cashew** producer among the options: **Maharashtra** 1× 2013
- **Coffee Board-Hyderabad** is wrong; it is in **Bengaluru** 1× 2023
- **National Horticulture Board**: **1984** 1× 2014

Board	Headquarters
Tea Board	Kolkata
Coffee Board	Bengaluru
Rubber Board	Kottayam
Spices Board	Kochi
Coconut Board	Kochi
Central Silk Board	Bengaluru
Tobacco Board	Guntur
National Horticulture Board	Gurugram (1984)

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TRAP ALERT

- Leaders that **changed**: coconut (Kerala to **Karnataka**), tobacco (AP to **Gujarat**), world tea (India to **China**).
- Leaders that **did not change**: coffee **Karnataka**, rubber **Kerala**, tea **Assam**, mulberry silk **Karnataka**, muga **Assam**.

UP LINK

UP is not a plantation state. It appears in tobacco questions (an old key named UP) and in silk weaving: **Varanasi (Banarasi)** silk sarees use mulberry silk mostly from Karnataka.

LAST-MINUTE 10

1. **Assam** = tea, muga, eri; **Jharkhand** = tasar
2. **Karnataka** = coffee, mulberry silk, silk yarn and fabric
3. **Kerala** = rubber, cardamom, pepper; Garden of Spices
4. Tea = Green Gold, stimulant; 200 cm rain on slopes
5. Coconut: Kerala, now **Karnataka**
6. Tobacco: AP, now **Gujarat**
7. India leads world in **ginger**, turmeric, chilli
8. **Coffee Board = Bengaluru**; Rubber Board = Kottayam; Tea Board = Kolkata
9. National Horticulture Board **1984**
10. **Maharashtra** = major cashew producer

QUANTUM IAS & PCS

Chapter 19

Farming Practices, Horticulture and Livestock

Built from 67 UPPSC questions (1990–2024). Focus: crop-state matching, maize, seed classes, fruit varieties, livestock and milk.

ASKED MOST · REVISE FIRST

- **Crop-state matching**: **jute-West Bengal, tea-Assam, rubber-Kerala, sugarcane-UP, groundnut-AP/Gujarat, mustard-Rajasthan, soybean-MP** 7× 2008, 2008, 2010, 2011, 2014, 2016, 2017
- **Maize** is grown **all year round**; it is a **C4** plant; Shaktiman-I and II are maize 4× 2006, 2007, 2014, 2015
- **West Bengal** = jute industry and largest jute area 2× 2004, 2007
- **UP** = top in **milk**, foodgrains, potato (2013) 3× 2008, 2014, 2014
- **Amrapali** = **Dasheri × Neelam** mango 2× 2013, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

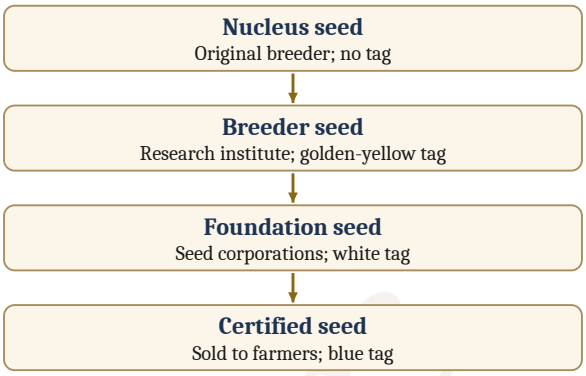
19.1 Shifting Cultivation

- **Jhum** is practised mostly in **Nagaland** 1× 2005
- Jhum is a problem in the hills of **Odisha and MP** (per the 1998 key) 1× 1998
- Match: **Western Ghats-Kumari, SE Rajasthan-Waltre, North-east-Jhum, MP and Chhattisgarh-Dahiya** 1× 2017

19.2 Seeds, Diseases and Weeds

- Seed stages: **Nucleus, Breeder, Foundation, Certified** 1× 2023
- **Certified** seed tag: **blue** 1× 2016
- Match: **sugarcane-red rot, paddy-khaira, arhar-wilt, potato-late blight** 1× 2014
- **Rice-green hair** is wrong 1× 2015
- **Paddy-bathua** is wrong (bathua is a wheat weed) 1× 2016

SEED MULTIPLICATION CHAIN



19.3 Fibre Crops

- Jute industry: **West Bengal** {2004}; largest jute area: **West Bengal** 1× 2007
- India was the largest jute producer (Bangladesh now leads); **Yellow Revolution** is about oilseeds, not jute (*) 1× 2023
- Jute is **native to India** and a **kharif** crop (both true, R not the explanation) 1× 2024
- Fibre crops: **cotton, jute, sunhemp** 1× 2017
- Cotton is a **kharif** crop needing **50-75 cm** rain (both true) 1× 2024

19.4 Maize, Saffron and Cropping

- Grown **all year round**: **maize** 2× 2007, 2015
- **C4 plant**: **maize** 1× 2014
- Maize (then): **Karnataka, Bihar, Rajasthan, Andhra Pradesh** (descending) 1× 2010
- **Shaktiman-I and II**: **maize** (quality protein) 1× 2006
- **Saffron**: **Jammu and Kashmir** (Kashmir, Pampore) 2× 1990, 2007
- Best crop cycle for **eastern UP**: **paddy-maize-wheat** 1× 2008
- Subsistence crops: **coarse cereals and rice** 1× 2007
- Cereal product: **oatmeal** 1× 2002
- First fully **organic** state: **Sikkim** 1× 2016

19.5 Leading States and Varieties

- **Banana-Maharashtra** was the wrong pair then (Tamil Nadu led); now Maharashtra leads 2× 2008, 2011
- Largest potato producer (2013): **UP** {2014}; "potato-MP" is wrong 1× 2016
- Largest share of **foodgrains**: **UP** 1× 2008
- **Gujarat** leads in **bajra** (with Rajasthan) 1× 2008
- Match: **groundnut-AP, mustard-Rajasthan, soybean-MP, coconut-Kerala** 1× 2008
- Match: **jute-WB, tea-Assam, rubber-Kerala, sugarcane-UP** 3× 2010, 2014, 2017

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- Season: **sunflower-all seasons, muskmelon-zaid, cotton-kharif, linseed-rabi** 1× 2016
- **Paddy-Radhey** is wrong (Radhey is a gram variety) 1× 2016
- **Wheat-Shusk Samrat** is wrong (it is a pea/lentil variety, not wheat) 1× 2016
- Best potato for processing: **Kufri Chipsona-2** 1× 2008

19.6 Horticulture

- India is the largest producer of **mango and papaya** (2020-21) 1× 2021
- India is **second** in vegetables {2006}; **China** is first 1× 2015
- India gives about **15%** of world fruit 1× 2013
- Top fruit in India by output: **banana** {2023}; top banana state: **Maharashtra** (per key) 1× 2013
- National Research Centre for Banana: **Trichy** 1× 2014
- Seedless mango: **Sindhu** 1× 2007
- **Amrapali** = Dasheri × Neelam {2013, 2016}; regular-bearing variety: none of the options 1× 2014
- **Lalit**: **guava** {2008}; **Kanchan**: **aonla** 1× 2015
- "Ganga Varanasi" is not a listed variety (*) 1× 2015
- **Sultana, Gulabi, Kali Champa**: **grapes** 1× 2018

Variety	Crop
Amrapali, Mallika, Sindhu	Mango
Lalit, Allahabad Safeda	Guava
Kanchan, Banarasi	Aonla
Sultana, Gulabi	Grapes
Kufri Chipsona	Potato
Shaktiman	Maize
Radhey	Gram

19.7 Mixed Statements

- Wrong: cotton fibre length 1-5 cm 1× 2013
- Wrong: linseed grows all year 1× 2013
- Wrong: Bihar is the largest lac producer (Jharkhand is) 1× 2013
- Wrong: **Aman** rice is sown in March-April (it is June-July) 1× 2013
- Rajma is the poor nitrogen fixer 1× 2013
- More than one statement untrue (Kaushal is groundnut; MP leads in pulses) (*) 1× 2013
- India is the largest producer of none of sugarcane, tobacco, coffee, oilseeds (*) 1× 2010

19.8 Livestock, Milk and Fish

- Stock farming = **breeding animals** 1× 2005
- About one-third of cattle: **West Bengal, UP, MP** (2019 census) (*) 2× 2006, 2009
- **UP** has the most **buffaloes**; WB the most cattle; Telangana the most sheep (*) 1× 2008
- High-milk cow breed: **Sahiwal** 1× 2004
- India is **first** in milk {2008}; top milk state (2011-12): **UP** 1× 2014
- **White Revolution** = **milk** 1× 2015
- West Bengal led in fish (mostly **inland**), now **Andhra Pradesh** leads (*) 2× 2007, 2015

Livestock (2019 census)	Leading state
Cattle	West Bengal
Buffalo	Uttar Pradesh
Sheep	Telangana
Goat	Rajasthan
Pig	Assam
Poultry	Tamil Nadu
Total livestock	Uttar Pradesh

UP LINK

UP leads India in **milk, buffaloes, total livestock, foodgrains, potato, sugarcane and wheat**. Best eastern-UP cycle: **paddy-maize-wheat**. UP cattle breeds: **Ponwar**, **Gangatiri**; buffalo: **Bhadawari**. Guava **Allahabad Safeda** and **Lalit** come from UP.

LAST-MINUTE 10

1. Jhum = **Nagaland**; Kumari = Western Ghats; Dahiya = MP
2. Seeds: nucleus, breeder, foundation, **certified (blue tag)**
3. **Maize** = all year, C4, Shaktiman
4. Saffron = **Kashmir**; first organic state = **Sikkim**
5. Jute = **West Bengal**; native to India; kharif
6. **UP** = milk, buffalo, foodgrains, potato
7. **Sahiwal** = top milk cow; India first in milk
8. Banana = top fruit; NRC Banana = **Trichy**
9. **Amrapali** = Dasheri × Neelam; **Lalit** = guava; **Kanchan** = aonla
10. Fish: **Andhra Pradesh** now leads

Chapter 20

Mineral Resources I: Metallic and Non-metallic Minerals

Built from 60 UPPSC questions (1991–2025). Focus: mine-mineral and mine-state matching, Dharwar rocks, iron ore, mica, uranium.

ASKED MOST · REVISE FIRST

- **Mine-mineral matching** is the chapter's main format: **Khetri-copper, Kodarma-mica, Dalli-Rajhara-iron, Ankleshwar-oil, Zawar-zinc/lead/silver, Jaduguda-uranium, Talcher-coal, Kolar-gold** 13× 1998, 2001, 2002, 2005, 2008, 2008, 2008, 2009, 2012, 2013, 2016, 2016, 2020
- **Dharwar** rocks are the most mineralised and hold **iron ore** 7× 1994, 1995, 2005, 2012, 2014, 2017, 2019
- Iron-ore mines and states: **Gorumahisani, Badampahar-Odisha; Bailadila, Dalli-Rajhara-Chhattisgarh; Noamundi-Jharkhand; Kudremukh-Karnataka** 4× 2004, 2010, 2015, 2025
- **Mica**: Rajasthan now (AP earlier); MP does not produce it; belt in **Hazaribagh, Gaya, Munger** 5× 2001, 2003, 2004, 2009, 2010
- **Rajasthan**: near monopoly in **zinc**; also limestone, gypsum, salt; largest copper reserves 3× 2011, 2013, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

20.1 Rock Systems

- Most mineralised rock system: **Dharwar** 3× 1995, 2005, 2017
- Iron ore occurs in **Dharwar** rocks 4× 1994, 2012, 2014, 2019
- **Vindhyan** rocks are rich in **limestone** 1× 2004
- **Gondwana** = about 95% of coal; ferrous and non-ferrous minerals are in Dharwar (both true, R not the explanation) 1× 2017

20.2 Critical Minerals

- **Lithium** inferred resources at **Reasi** (J&K); India imports lithium, nickel, cobalt (both true) 1× 2023

20.3 Iron Ore

- Iron-ore regions: **Kudremukh and Singhbhum** (*) 1× 1992
- Match: **Gurumahisani-Odisha, Bailadila-Chhattisgarh, Noamundi-Jharkhand, Kudremukh-Karnataka** 1× 2025
- Match: **Badam Pahar-Odisha, Dalli-Rajhara-Chhattisgarh, Kudremukh-Karnataka, Noamundi-Jharkhand** 1× 2010
- **Bailadila-MP** is wrong; it is in **Chhattisgarh** 1× 2004
- Largest mechanised mine: **Bailadila** 1× 2015
- Top iron-ore states (2020-21): **Odisha and Chhattisgarh** 1× 2022
- True: richest deposits in **Karnataka** (per key); **Odisha** is the largest producer 1× 2010
- **Iron ore-Guntur** is wrong 1× 2017

Iron-ore mine	State
Gorumahisani, Badampahar, Keonjhar	Odisha
Noamundi, Gua, Chiria	Jharkhand
Bailadila, Dalli-Rajhara	Chhattisgarh
Kudremukh, Bellary	Karnataka
North and South Goa	Goa

20.4 Manganese, Chromite, Nickel

- Leading **manganese** producer: **Maharashtra** (MP for many years; Odisha has most reserves) 2× 2008, 2012
- **Srikakulam** is **not** in Karnataka (it is in AP) 1× 2023
- Monopoly in **chromite**: **Odisha** (Sukinda) 2× 2008, 2019

20.5 Copper, Zinc, Bauxite, Tin

- Largest **copper** reserves: **Rajasthan** 1× 2011
- Match: **Chandrapur-Maharashtra, Hassan-Karnataka, Khammam-AP, Khetri-Rajasthan** 1× 2011
- Near monopoly in **zinc**: **Rajasthan** 1× 2013
- **Bauxite** = ore of **aluminium** 1× 1997
- Top bauxite states: **Odisha and Gujarat** 2× 2021, one-liner
- **Tin**: reserves and production in **Chhattisgarh** (Dantewada) 2× 2009, 2015

20.6 Mica, Asbestos and Stones

- Largest mica producer: **Rajasthan** now (Andhra Pradesh in 2001-04) 2× 2001, 2004
- **MP** does **not** produce mica 2× 2003, 2010
- Largest mica belt: **Hazaribagh, Gaya, Munger** 1× 2009
- India led the world in **sheet mica** in 2001 (*) 1× 2001
- Asbestos: no state produces it now (*) 1× 2015
- Self-sufficiency in 1991: **graphite** (*) 1× 1991
- Best **marble**: **Makrana** (Rajasthan) {1992}; marble is a **metamorphic** rock 1× 1991
- **Garnet** is linked with metamorphic rock 1× 2010
- **Granite** slabs and slates: **Lalitpur** (UP) 1× 2002
- Sonbhadra minerals: **andalusite, pyrite, dolomite** (not uranium) 1× 2001

20.7 Atomic Minerals, Gold, Diamonds

- **Monazite** is abundant in **Kerala** {2013}; also Tamil Nadu 1× 2002
- **Jaduguda** (Jharkhand): **uranium** 1× 2007

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- Match: **Domiasiat-Meghalaya, Lambapur-AP (Telangana), Rohil-Rajasthan, Gogi-Karnataka** 1× 2008
- Diamond mines: **Madhya Pradesh** (Panna) 1× 2014

20.8 Mixed Mine Matching

- World: **silver-Mexico, mica-India, iron ore-Russia, copper-Chile** 1× 2017
- **Manganese-MP, nickel-Odisha, lead-zinc-Rajasthan, asbestos-AP** 1× 1998
- **Iron ore-Odisha, copper-Rajasthan, gold-Karnataka, mica-AP** 1× 2008
- **Mineral oil-Gujarat, gypsum-Rajasthan, gold-Karnataka, bauxite-Odisha** 2× 2006, 2010
- **Makum-coal, Dalli-Rajhara-iron, Koraput-bauxite, Chitradurga-manganese** 1× 2016
- **Kamptee-copper** is wrong (Kamptee = coal) 1× 2009
- **Badam Pahar-iron, Kodarma-mica, Mosabani-copper, Rava-petroleum** 1× 2008
- **Nellore-manganese** is wrong (Nellore = mica) 1× 2002
- **Ankleshwar-petroleum, Dalli-Rajhara-iron, Kodarma-mica, Khetri-copper** 1× 2001
- **Iron-Keonjhar, copper-Khetri, coal-Korba, mica-Kodarma** 1× 2005
- **Gold-Kolar, coal-Jharia, copper-Khetri, iron-Kudremukh** 1× 2008
- **Coal-Talcher, copper-Khetri, iron-Kudremukh, silver-Zawar** 1× 1998
- **Manganese-Koraput** is wrong (Koraput = bauxite) 1× 2013
- **Copper-Khetri, gas plant-Auraiya, aluminium-Korba, petroleum-Kochi** 1× 2002
- **Palamu-bauxite, Hazaribagh-mica, Khetri-copper, Keonjhar-manganese** 1× 2008
- **Gurumahisani-iron, Talcher-coal, Jaduguda-uranium, Zawar-lead** 2× 2012, 2016
- Wrong: **copper-Chitradurga** and **manganese-Bhilwara** 1× 2020
- **Limestone-cement, copper-electrical, bauxite-aircraft, manganese-steel** 1× 2021
- **Rajasthan**: limestone, gypsum, salt, zinc 1× 2023

Place	Mineral
Khetri	Copper
Kodarma, Hazaribagh	Mica
Zawar	Zinc, lead, silver
Jaduguda	Uranium
Kolar, Hutti	Gold
Panna	Diamond
Makrana	Marble
Koraput, Panchpatmali	Bauxite
Talcher, Jharia, Korba	Coal
Ankleshwar	Petroleum
Sukinda	Chromite

TRAP ALERT

- Distractor places: **Guntur** (tobacco), **Nellore** (mica), **Kamptee** (coal), **Koraput** (bauxite), **Bhilwara** (zinc, not manganese).
- **Bailadila** is now in **Chhattisgarh**, not MP.

UP LINK

UP's minerals are in the south: **Sonbhadra** (coal of the Singrauli field, limestone at Churk-Dalla-Obra, andalusite, pyrite, dolomite), **Lalitpur** (granite, rock phosphate), **Mirzapur** (Chunar sandstone) and **Shankargarh** (Prayagraj, silica sand for glass).

LAST-MINUTE 10

1. **Dharwar** = most minerals, iron ore; **Vindhyan** = limestone
2. **Odisha** = iron, chromite, bauxite leader
3. **Bailadila** = Chhattisgarh, largest mechanised mine
4. **Rajasthan** = zinc, copper reserves, gypsum, marble, mica now
5. **Khetri** = copper; **Kodarma** = mica; **Zawar** = zinc-silver
6. **Jaduguda** = uranium; monazite = Kerala beaches
7. **Panna** = diamond; **Makrana** = marble
8. Tin = **Chhattisgarh**; Lithium = **Reasi**
9. Bauxite = aluminium; Odisha and Gujarat lead
10. **Lalitpur** = granite; **Sonbhadra** = andalusite, pyrite

Chapter 21

Mineral Resources II: Coal, Petroleum and Natural Gas

Built from 59 UPPSC questions (1998–2025). Focus: coalfield-state and coalfield-valley matching, reserves vs production, oilfields, refinery-state matching, natural gas.

ASKED MOST · REVISE FIRST

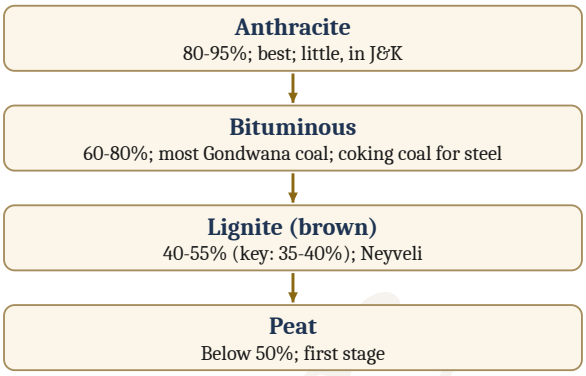
- **Coalfield matching** is the main format: **Talcher-Odisha, Singareni-Telangana, Korba/Bisrampur-Chhattisgarh, Sohagpur/Umaria-MP, Jharia/Karanpura-Jharkhand, Raniganj-WB, Namchik-Arunachal, Makum-Assam** 10× 2008, 2010, 2010, 2016, 2016, 2017, 2019, 2023, 2024, 2025
- **Refinery matching**: **Numaligarh/Noonmati-Assam, Koyali/Jamnagar-Gujarat, Tatipaka-AP, Haldia-WB, Barauni-Bihar, Manali/Nagapattinam-TN** 7× 2005, 2008, 2010, 2013, 2014, 2016, 2025
- **Gondwana** rocks give over **90%** of India's coal 4× 2006, 2008, 2016, 2017
- Largest coal **producer**: **Odisha** (then Chhattisgarh) 5× 2002, 2009, 2009, 2010, 2014
- **Gujarat** = leading onshore petroleum state; **Bombay High** gives most natural gas 5× 2003, 2004, 2010, 2012, 2013

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

21.1 Coal: Reserves and Production

- Main source of coal: **Gondwana** system {2006, 2008}; over **90%** of coal 2× 2016, 2017
- India is a leading coal producer with vast reserves in **Jharkhand and Chhattisgarh** (both true) 1× 2023
- Over half of proved reserves in **Jharkhand and Odisha**; key says both true, R not the explanation (resources are now about 361 bt, not 293.5) 1× 2019
- Reserves order in 2014: Jharkhand first; **Odisha** leads in total resources now (*) 1× 2014
- Key's order **Chhattisgarh, Jharkhand, Odisha, Maharashtra** fits **production**; proved reserves (2014) = Jharkhand, Odisha, Chhattisgarh, Maharashtra 1× 2016
- Two leading producers: **Chhattisgarh and Odisha** 1× 2010
- Production order 2008: Jharkhand, Chhattisgarh, Odisha, MP; now **Odisha, Chhattisgarh, MP, Jharkhand** (*) 1× 2008
- Largest coal producer: **Odisha** 3× 2002, 2009, 2014
- Top three producers: **Odisha, Chhattisgarh, MP** 1× 2009

COAL GRADES BY CARBON CONTENT



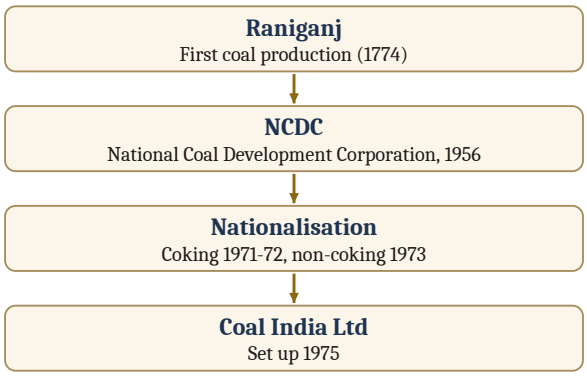
21.2 Coalfields and their States

- **Chota Nagpur** industry grew around **coal** {2004}; it gave about **60%** of coal (then) 1× 1998
- Valleys: **Damodar-Barakar, Son-Umaria, Godavari-Singareni, Mahanadi-Talcher** 1× 2010
- Valleys: **Talcher-Mahanadi, Karanpura-Damodar, Singrauli-Son, Singareni-Godavari** 1× 2019
- **Talcher**: **Odisha** 1× 2010
- **Singareni-MP** is wrong (it is **Telangana**) 1× 2023
- Correct: **Raniganj-WB, Ramgarh-Jharkhand, Talcher-Odisha**; **Maolong** is not Arunachal (it is in Meghalaya) 1× 2025
- **Hutar-Jharkhand, Ib River-Odisha, Tatapani-Ramkola-Chhattisgarh, Umaria-MP** 1× 2024
- **Karanpura-Jharkhand, Singareni-AP (now Telangana), Neyveli-TN, Korba-Chhattisgarh** 1× 2010
- **Rajmahal-Jharkhand, Sohagpur-MP, Bisrampur-Chhattisgarh, Raniganj-WB** 1× 2017
- **Makum-Assam, Namchik-Arunachal, Giridih-Jharkhand, Sohagpur-MP** 1× 2019
- **Namchik-Namphuk**: **Arunachal Pradesh** 1× 2008
- **Jhilmili-Chhattisgarh, Karanpura-Jharkhand, Rampur Himgir-Odisha, Sohagpur-MP** 1× 2016
- **Rangit Valley-Sikkim, Tandur-Telangana, Warora-Maharashtra, Lalmatia-Jharkhand** 1× 2016
- **Bisrampur** is known for **coal** 2× 2004, 2008
- Coalfield with the largest reserves: **Raniganj** 1× 1999
- **Jharia**: **Jharkhand** (best coking coal) 1× 2014

Coalfield	State	Valley
Raniganj	West Bengal	Damodar
Jharia, Bokaro, Giridih, Karanpura	Jharkhand	Damodar
Rajmahal (Lalmatia), Hutar	Jharkhand	Rajmahal, North Koel
Talcher, Ib River, Rampur-Himgir	Odisha	Mahanadi

Coalfield	State	Valley
Korba, Bisrampur, Jhilmili, Tatapani	Chhattisgarh	Mahanadi, Son
Singrauli, Sohagpur, Umaria	Madhya Pradesh	Son
Singareni, Tandur	Telangana	Godavari
Wardha, Kamptee, Warora	Maharashtra	Wardha
Makum; Namchik-Namphuk	Assam; Arunachal	Tertiary coal

COAL HISTORY IN ORDER



- Order: **Raniganj, NCDC, nationalisation, CIL** 1× 2023

21.3 Lignite and the Use of Coal

- Lignite has about 35-40% carbon (true); **Jharkhand** is **not** the lignite leader (A true, R false) 1× 2002
- Largest lignite deposit: **Tamil Nadu** {2013}; largest lignite field: **Neyveli** 1× 2019
- **Jharkhand** does **not** have large lignite reserves 1× 2017
- Most coal is used for **power (energy) generation** 1× 2008
- Coal is a major part of inter-state rail traffic; Bengal-Jharkhand coal feeds the north-west (R explains A) 1× 2010

21.4 Petroleum: Oilfields

- First oil well: **Makum** (1867) per the original key; revised key **Digboi** (oil struck 1889, refinery 1901) 1× 2013
- Leading petroleum producer (onshore): **Gujarat** 3× 2003, 2004, 2010
- **Ankleshwar**: **petroleum** 1× 2004
- **Navgam** oilfield: **Gujarat** 1× 2013
- Wrong fact: Bombay High is **not** 60 km off Mumbai (it is about 176 km) 1× 2014

Region	Oil and gas fields
Assam	Digboi, Naharkatiya, Moran, Lakwa, Rudrasagar
Gujarat	Ankleshwar, Kalol, Navagam, Mehsana, Lunej
Rajasthan (Barmer)	Mangala, Bhagyam, Aishwariya
Offshore Mumbai	Mumbai High, Bassein (gas), Panna-Mukta
KG basin	KG-D6 (gas), Ravva

21.5 Refineries

- Biggest refinery: **Jamnagar** (Gujarat) 1× 2016
- **Hatia** has no refinery (it is the Ranchi heavy-engineering town) 1× 2010
- **Noonmati** (Guwahati): **Assam** 1× 2008
- **Panagudi-Tamil Nadu** is the correct pair 1× 2006
- **Nayara-Gujarat, Manali-TN, Numaligarh-Assam, Tatipaka-AP** 1× 2025
- **Haldia-WB, Jamnagar-Gujarat, Kochi-Kerala, Numaligarh-Assam** 1× 2005
- **Numaligarh-Assam, Tatipaka-AP, Koyali-Gujarat, Barauni-Bihar** 1× 2010
- **Tatipaka**: **Andhra Pradesh** 1× 2016
- **Noonmati-Assam, Koyali-Gujarat, Mangalore-Karnataka, Panipat-Haryana** 1× 2014
- **Tatipaka-AP, Koyali-Gujarat, Nagapattinam-TN, Numaligarh-Assam** 1× 2013
- **Koyali**: **Gujarat** (near Vadodara) 1× 2016

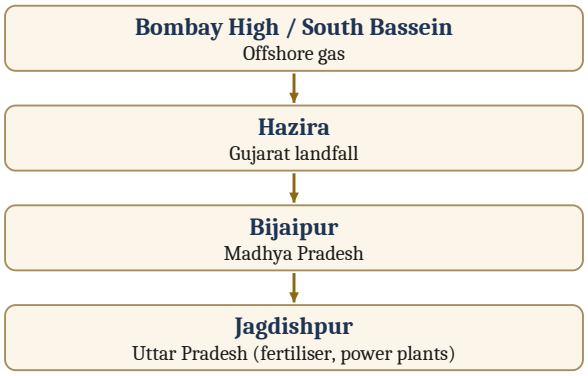
Refinery	State
Digboi (1901), Noonmati, Bongaigaon, Numaligarh	Assam
Barauni	Bihar
Haldia	West Bengal
Koyali, Jamnagar, Vadinar (Nayara)	Gujarat
Mathura	Uttar Pradesh
Panipat	Haryana
Mangaluru	Karnataka
Kochi	Kerala
Manali, Nagapattinam	Tamil Nadu
Visakhapatnam, Tatipaka	Andhra Pradesh

21.6 Strategic Reserves and Natural Gas

- Strategic reserves at **Visakhapatnam, Mangaluru, Padur**; **Ankleshwar** not chosen 1× 2008
- "Hydrocarbon Vision 2025" (key: "Hydrogen") = **storage/warehousing of petroleum** 1× 2004
- Oil exploration: **Oil India Ltd** (with ONGC) 1× 2004
- **HBJ** pipeline carries gas from the **South Basin** (Bassein) {2006}; built by **GAIL** 1× 2014
- Reliance's huge gas find: **coastal Andhra** (KG basin) 1× 2002
- **KG-D6**: **gas** 1× 2011
- Most natural gas: **Bombay High** 2× 2012, 2013

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HBJ GAS PIPELINE (GAIL, 1987)



TRAP ALERT

- Coal "first" depends on the word: **resources** = Odisha; **production** = Odisha; **coking coal** = Jharkhand (Jharia); **field with the largest reserves** = Raniganj.
- **Hatia** is not a refinery; **Haldia** is West Bengal, not Odisha; **Singareni** is Telangana, not MP; **Maolong (Mawlong)** is Meghalaya.
- **Ankleshwar** is an oilfield, not a strategic reserve site.

UP LINK

UP has the **Mathura** refinery (IOC, 1982), the **Singrauli** coalfield's UP part in **Sonbhadra** (feeding Anpara, Obra, Rihand, Singrauli power plants), and the end of the **HBJ pipeline at Jagdishpur** (Amethi), which feeds gas plants such as **Auraiya** and fertiliser units in UP.

LAST-MINUTE 10

1. **Gondwana** = over 90% of coal; Tertiary coal = Assam, Arunachal
2. Coal production: **Odisha, Chhattisgarh**; best coking coal: **Jharia**
3. **Raniganj** = first coal (1774), largest reserves
4. Order: Raniganj, NCDC, nationalisation, **CIL 1975**
5. Lignite = **Neyveli** (Tamil Nadu); not Jharkhand
6. First well **Makum 1867**; Digboi oil 1889, refinery 1901
7. **Gujarat** = onshore oil; **Bombay High** = most gas
8. **Jamnagar** = biggest refinery; **Koyali** = Gujarat; **Tatipaka** = AP
9. **Hatia** = no refinery; **Noonmati, Numaligarh** = Assam
10. HBJ = **GAIL**, Hazira to **Jagdishpur**; KG-D6 = gas

Chapter 22

Power and Energy Resources

Built from 38 UPPSC questions (1991–2023). Focus: energy mix, power-plant-state matching, nuclear stations, wind, solar, tidal and hot springs.

ASKED MOST · REVISE FIRST

- **Thermal (coal)** is the largest source of energy and of power generation 4× 2002, 2006, 2008, 2011
- **Nuclear station-state matching**: **Narora-UP, Rawatbhata (Kota)-Rajasthan, Kakrapar-Gujarat, Kaiga-Karnataka, Tarapur-Maharashtra** 5× 1999, 2002, 2007, 2015, 2017
- **Thermal plant-state matching**: **Ukai-Gujarat, Patratu-Jharkhand, Kahalgaon-Bihar, Korba-Chhattisgarh, Kudgi-Karnataka** 5× 2006, 2010, 2013, 2016, 2023
- India is self-sufficient in **thorium** 3× 2002, 2003, 2005
- **Gujarat** = first in **wind** power and **tidal** potential (Khambhat, Kachchh) 3× 1991, 2013, 2022

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

22.1 The Energy Mix

- Largest source of energy: **coal** 2× 2002, 2008
- Largest share in power generation: **thermal** 2× 2006, 2011
- Order in the 2000s: thermal, hydro, nuclear, wind; now wind has overtaken hydro and nuclear (*) 2× 2010, 2014
- Installed capacity (2019): **hydro, gas, nuclear, diesel** 1× 2016
- **Biogas** is **not** a commercial source 1× 2013

Source (end 2025)	Capacity	Share
Coal and lignite	226 GW	44%
Solar	136 GW	26%
Wind	55 GW	11%
Large hydro	51 GW	10%
Gas and diesel	21 GW	4%
Nuclear	8.8 GW	2%

22.2 Thermal Power Stations

- **Singrauli**: NTPC's **oldest** unit; it is in **UP** (Sonbhadra), not Chhattisgarh 1× 2021
- **Obra** thermal plant was built with **Russian** (Soviet) help 1× 2008
- **Ukai-Gujarat, Patratu-Jharkhand, Pench-MP, Dabhol-Maharashtra** 2× 2010, 2013
- **Uran** gas plant: **Maharashtra** 1× 2010
- **Ramagundam**: AP then, **Telangana** now 1× 2004
- **Kahalgaon-Bihar, Khargone-MP, Korba-Chhattisgarh, Kudgi-Karnataka** 1× 2023
- **Paras-AP** is wrong (Paras is in **Maharashtra**) 1× 2006
- Thermal capacity rank 2008: **Maharashtra** first; now **UP** leads (*) 1× 2008
- Plant Load Factor then: **65-70%** 1× 2004
- **Korba-wind** is wrong (Korba = thermal) 1× 2016
- **Chamera-hydro, Gwal Pahari-solar, Kudankulam-atomic, Patratu-thermal** 1× 2019

22.3 Nuclear Power

- Self-sufficient in **thorium** 3× 2002, 2003, 2005
- Nuclear energy is promising, but nuclear minerals are **not** everywhere (A true, R false) 1× 2006
- First nuclear reactor: **Apsara** (1956, Trombay) 1× one-liner
- **Kota-Rajasthan, Tarapur-Maharashtra, Kakrapar-Gujarat, Narora-UP** 1× 1999
- **Gujarat-Kakrapar, Karnataka-Kaiga, Rajasthan-Rawatbhata, UP-Narora** 1× 2002
- Commissioning-year pairs: none correct (*) 1× 2007
- **Ennore-Meghalaya** is wrong (Ennore = Tamil Nadu, thermal) 1× 2015
- **Mithi Virdi** was planned with the **USA** (moved to **Kovvada**, AP) 1× 2014
- **Haripur-Haryana** is wrong; Haripur is **West Bengal**; Haryana's is **Gorakhpur** 1× 2017
- **Anushakti Vidyut Nigam** = NPCIL + **NTPC** 1× 2015

INDIA'S THREE-STAGE NUCLEAR PROGRAMME

Stage 1: PHWR
Natural uranium; produces plutonium

↓

Stage 2: Fast breeder
Plutonium; PFBR at Kalpakkam

↓

Stage 3: Thorium reactors
Uses India's vast thorium (monazite)

Nuclear station	State	First unit
Tarapur	Maharashtra	1969
Rawatbhata (Kota)	Rajasthan	1973
Kalpakkam	Tamil Nadu	1984
Narora	Uttar Pradesh	1991
Kakrapar	Gujarat	1993
Kaiga	Karnataka	2000

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Nuclear station	State	First unit
Kudankulam	Tamil Nadu	2013 (Russia)
Gorakhpur	Haryana	Under construction

22.4 Hydro, Wind, Solar and Tidal

- Hydro station: **Koyna** (Maharashtra) 1× 1992
- Correct: **Sabarigiri-hydro, Ghatprabha-irrigation, Ramganga-multipurpose**; Idukki is **hydro**, not thermal 1× 2017
- **Shivanasamudra** = hydel (1902); **Kalpakkam** = nuclear 1× 2023
- First in wind power: **Gujarat** (Tamil Nadu earlier) 1× 2013
- Asia's biggest 150 MW wind project (1999): **Tamil Nadu** (Muppandal) 1× 1999
- First in installed capacity: **Gujarat** per key (Maharashtra in 2006); Gujarat leads now with renewables 1× 2006
- **Rampura**, first solar-powered village: **UP** 1× 2010
- First UT on 100% solar: **Diu** 1× 2017
- Largest tidal energy area: **Khambhat** coast {2022}; most potential: **Bhavnagar** (Gulf of Khambhat) 1× 1991

22.5 Hot Springs

- **Lasundra-Rajasthan** is wrong (Lasundra is in **Gujarat**) 1× 2014
- **Jwalamukhi-J&K** is wrong (it is in **Himachal Pradesh**) 1× 2015

TRAP ALERT

- **Singrauli** NTPC plant is in **UP**; **Paras** is Maharashtra; **Ennore** is Tamil Nadu; **Haripur** is West Bengal.
- **Idukki, Koyna, Sabarigiri, Shivanasamudra** = hydro; **Korba, Patratu, Obra** = thermal.
- Lasundra (Gujarat), Jwalamukhi (HP), Manikaran (HP), Puga (Ladakh): geothermal sites.

UP LINK

UP has **Narora** atomic plant (Bulandshahr, 1991), **Singrauli** (NTPC's oldest), **Obra** (Soviet aid), **Anpara**, **Rihand** in Sonbhadra (India's "power capital"), the gas plant at **Auraiya**, and **Rampura**, India's first solar village. UP now leads in thermal capacity.

LAST-MINUTE 10

1. Coal/thermal = largest energy and power source
2. India self-sufficient in **thorium**; first reactor **Apsara**
3. **Narora-UP, Kakrapar-Gujarat, Kaiga-Karnataka, Rawatbhata-Rajasthan**
4. Haripur = WB; Gorakhpur nuclear = **Haryana**; Mithi Virdi to Kovvada (USA)
5. Singrauli = NTPC's oldest (UP); **Obra** = Russia
6. Ukai = Gujarat; Patratu = Jharkhand; Kahalgaon = Bihar; Kudgi = Karnataka
7. **Gujarat** = wind, tidal (Khambhat), largest capacity now
8. **Rampura** (UP) = first solar village; **Diu** = first solar UT
9. Shivanasamudra = hydel; Idukki = hydro
10. Jwalamukhi = HP; Lasundra = Gujarat

Chapter 23

Industries and Industrial Centres

Built from 42 UPPSC questions (1994–2024). Focus: steel plants, aluminium centres, PSU headquarters, textile and agro centres, fertilizer, paper and cement towns.

ASKED MOST · REVISE FIRST

- **Place-industry matching** dominates: **Renukoot-aluminium, Korba-BALCO, Muri-aluminium (Jharkhand), Sindri-fertilizer, Phulpur-fertilizer, Titagarh-paper, Bhadohi-carpet, Chunar-cement** 13× 1994, 1998, 2001, 2002, 2003, 2006, 2008, 2008, 2010, 2010, 2013, 2017, 2024
- **Wrong-pair** traps: **Bhilai-MP, BALCO-Raipur, Muri-Chhattisgarh, Amlai-Chhattisgarh, Dharmapuri-Odisha, Sindri-copper, Pipri-pesticides** 8× 2002, 2003, 2004, 2004, 2009, 2010, 2010, 2024
- **Gujarat** = largest petrochemical centre (**Jamnagar**) 2× 2006, 2013
- **UP**: most paper mills; **Phulpur** = biggest cooperative fertilizer plant; **Chunar** = cement 3× 2008, 2008, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

23.1 Iron and Steel

- **Bhilai-MP** is wrong now (it is in **Chhattisgarh**) 1× 2004
- **Rourkela** gets iron ore from **Keonjhar** 1× 2005
- Plant far from coalfields: **Bhadravati** (charcoal, hydel) 1× 2013
- **SAIL** was formed in **1973** (not in the options, *) 1× 2006

Steel plant	State	Helped by
TISCO, Jamshedpur (1907)	Jharkhand	Tata (private)
Bhadravati (1923)	Karnataka	Mysore State
Bhilai	Chhattisgarh	Soviet Union
Rourkela	Odisha	West Germany
Durgapur	West Bengal	Britain
Bokaro	Jharkhand	Soviet Union
Visakhapatnam (RINL)	Andhra Pradesh	Indigenous, shore-based

23.2 Aluminium and Copper

- **Korba** (Chhattisgarh): **aluminium** (BALCO) 2× 1994, 2001
- **BALCO-Raipur** is wrong (BALCO is at **Korba**) 2× 2002, 2003
- **BALCO-Korba, Hindalco-Renukoot, Indian Aluminium-Hirakud, NALCO-Koraput** 1× 2017
- **Alupuram-Kerala, Angul-Odisha, Belgaum-Karnataka, Korba-Chhattisgarh** 1× 2010
- **Muri-Chhattisgarh** is wrong (Muri is in **Jharkhand**) 1× 2024
- **Aluminium-J.K. Nagar, copper-Malanjkhand, zinc-Tundoo, jute-Bhatpara** 1× 2013

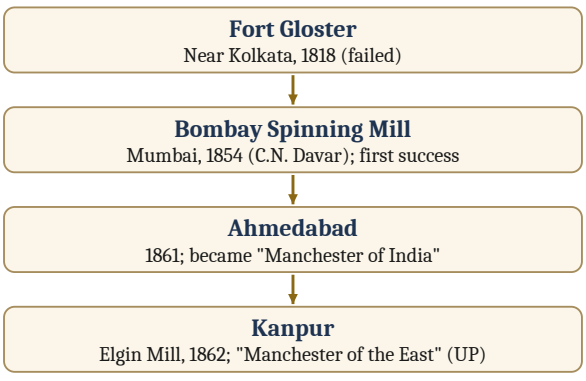
23.3 PSUs and Headquarters

- **ONGC-New Delhi, MECL-Nagpur, NMDC-Hyderabad, NALCO-Bhubaneswar** 1× 2021
- **SAIL-Delhi, Hindustan Zinc-Udaipur, HEC-Ranchi, Instrumentation Ltd-Kota** 1× 2010
- **Heavy engineering-Ranchi, machine tools-Pinjore, aluminium-Renukoot, fertilizer-Sindri** 1× 1998
- **BEL-Noida** is wrong (BEL's UP unit is at **Ghaziabad**) 1× 2006

23.4 Textiles, Carpets, Rubber

- Ancient industry of India: **cotton textile** 1× 2004
- First cotton mill (1818): **Fort Gloster**, near Kolkata 1× 2013
- **Ahmedabad** is not the largest cotton centre now (Mumbai historically; A false, R true) 1× 2020
- **Kakinada-jute, Virudhunagar-cotton textile, Channapatna-silk, Bhadohi-carpet** 2× 2010, 2013
- **Silk-Mysore, petrochemicals-Jawahar Nagar, fertilizer-Talcher, pharma-Rishikesh** 1× 2010
- First synthetic rubber plant: **Panipat** per key (many texts say **Bareilly**, 1963) 1× 2016
- Rubber industry at **Panjim** (per key) 1× 2007

FIRST COTTON MILLS



23.5 Fertilizers and Petrochemicals

- **Pipri-pesticides** is wrong (Pipri = Rihand hydel, Sonbhadra) 1× 2004
- **Sindri-copper** is wrong (Sindri = fertilizer) 1× 2010
- Biggest cooperative fertilizer plant: **Phulpur** (IFFCO, Prayagraj) 1× 2010
- **Visakhapatnam-shipbuilding, Muri-aluminium, Gurgaon-automobile, Panki-fertilizer** 1× 2006

- **Brajrajnagar-paper, Kaimur-cement, Haldia-petrochemical, Phulpur-fertilizer** 1× 2008
- Largest petrochemical factory: **Gujarat** {2006}; largest centre: **Jamnagar** 1× 2013
- **Nagapattinam-AP** is wrong (it is **Tamil Nadu**) 1× 2011
- **Dharmapuri-Odisha** is wrong (it is in **Tamil Nadu**) 1× 2010

23.6 Paper, Cement and Others

- **Amlai-Chhattisgarh** is wrong (Amlai paper mill is in **MP**, Shahdol) 1× 2009
- Most paper mills: **Uttar Pradesh** 1× 2008
- **Paper-Titagarh, cement-Lakheri, steel-Bhilai** 1× 2013
- **Chunar** (Mirzapur): **cement** 1× 2008
- India is **second** in world cement {2013}; largest cement state (2018-19): **Rajasthan** 1× 2016
- MP pesticide town: **Bhopal** 1× 2008
- **Sugar-Silvassa** is wrong 1× 2014
- **Pithampur** (MP): **automobile** hub 1× 2015
- **Diamond Parks**: zones to make and export diamonds and jewellery 1× 2004

Aluminium centre	State
Renukoot (Hindalco)	Uttar Pradesh
Korba (BALCO)	Chhattisgarh
Angul, Damanjodi (NALCO), Hirakud	Odisha
Muri	Jharkhand
J.K. Nagar	West Bengal
Belagavi	Karnataka
Alupuram	Kerala
Mettur	Tamil Nadu

23.7 Industrial Regions

- **Sivakasi** is **not** in Chota Nagpur {2015}; it is in the **Madurai-Coimbatore-Bengaluru** region 1× 2014

TRAP ALERT

- States changed: **Bhilai** (MP to Chhattisgarh), **Ramagundam** (AP to Telangana).
- **Muri** = Jharkhand; **Amlai** = MP; **Dharmapuri, Nagapattinam** = Tamil Nadu; **Sivakasi** = Tamil Nadu (matches, fireworks).
- Rourkela = Germany; Durgapur = Britain; Bhilai and Bokaro = Soviet Union.

UP LINK

UP centres asked: **Renukoot** (Hindalco aluminium), **Phulpur** (IFFCO), **Panki** (Kanpur, fertilizer), **Chunar** (cement), **Pipri** (Rihand hydel), **Bhadohi** (carpets), **Kanpur** (Manchester of the East), **Bareilly** (synthetic rubber, 1963). UP has the most paper mills. New: semiconductor unit at **Jewar**, mobiles at **Noida**, Defence Corridor nodes.

LAST-MINUTE 10

1. Bhilai = **Chhattisgarh** (Soviet); Rourkela = Germany; Durgapur = Britain
2. Rourkela ore from **Keonjhar**; Bhadravati far from coal
3. BALCO = **Korba**; Hindalco = **Renukoot**; NALCO = Koraput/Angul
4. **Muri** = Jharkhand; Alupuram = Kerala
5. HEC = **Ranchi**; Hindustan Zinc = **Udaipur**; NMDC = Hyderabad
6. First cotton mill **Fort Gloster 1818**
7. **Phulpur** = cooperative fertilizer; Sindri = fertilizer
8. **Jamnagar** = petrochemical capital; Gujarat leads
9. **UP** = most paper mills; **Chunar** = cement; **Rajasthan** = most cement
10. **Sivakasi** = Madurai-Coimbatore-Bengaluru region

Chapter 24

Research Centres and Institutes

Built from 34 UPPSC questions (1990–2021). Focus: ICAR institutes in UP, Hyderabad and Karnal institutes, institute-city matching, polar stations.

ASKED MOST · REVISE FIRST

- **UP institutes**: pulses **Kanpur**, sugarcane **Lucknow**, vegetables **Varanasi**, subtropical horticulture **Lucknow**, grassland and agroforestry **Jhansi** 11× 2004, 2010, 2010, 2010, 2013, 2015, 2015, 2016, 2017, 2017, 2021
- **Hyderabad**: **CRIDA**, ICRISAT, NAARM, rice directorate, NPPTI 6× 2008, 2010, 2013, 2013, 2014, 2015
- **Karnal**: **NDRI**, CSSRI, wheat research 3× 2013, 2013, 2016
- **Nagpur**: Indian Bureau of Mines, **NEERI** 2× 2010, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

24.1 Polar Research Stations

- Third Indian Antarctic station: **Bharati** (2012) 1× 2011

INDIA'S POLAR STATIONS

Dakshin Gangotri
Antarctica, 1983-84; first, now a supply base

↓

Maitri
Antarctica, 1989; Schirmacher Oasis

↓

Himadri
Arctic, 2008; Svalbard, Norway

↓

Bharati
Antarctica, 2012; Larsemann Hills

↓

Himansh
Himalaya (Spiti), 2016

24.2 ICAR Institutes in Uttar Pradesh

- Indian Institute of **Pulses** Research: **Kanpur** 3× 2004, 2010, 2015
- Indian Institute of **Vegetable** Research: **Varanasi** 2× 2016, 2017
- Central Institute of **Subtropical Horticulture**: **Lucknow** 1× 2010
- Indian Institute of **Sugarcane** Research: **Lucknow** 1× 2013
- **Subtropical Horticulture-Srinagar** is wrong (Srinagar has **Temperate** Horticulture) 1× 2017
- Agroforestry research centre: **Jhansi** {2010}; **Agroforestry-Banda** is wrong 1× 2021
- Indian **Grassland and Fodder** Research Institute: **Jhansi** 1× 2015
- **Goat** research: **Makhdoom** (Mathura) 1× 2017
- National Centre of **Organic** Farming: **Ghaziabad** 1× 2015
- Centre for **Social Forestry and Eco-Rehabilitation**: **Prayagraj** 1× 2019

UP city	Institute
Lucknow	Sugarcane Research; Subtropical Horticulture; NBRI; CDRI; CIMAP
Kanpur	Pulses Research; National Sugar Institute
Varanasi	Vegetable Research; IRRI South Asia centre
Jhansi	Grassland and Fodder; Agroforestry
Makhdoom (Mathura)	Goat research
Meerut (Modipuram)	Farming Systems Research
Ghaziabad	Organic and Natural Farming
Prayagraj	Eco-Rehabilitation (ICFRE)
Bareilly (Izatnagar)	IVRI; Avian research

24.3 Other Agricultural Institutes

- **ICRISAT**: **Hyderabad** (Patancheru) 1× 2014
- **CRIDA** (dryland agriculture): **Hyderabad** 3× 2008, 2013, 2015
- **NAARM**: **Hyderabad** 1× 2010
- National Institute of Agricultural Marketing: **Jaipur** 1× 2013
- Central Institute of Agricultural Engineering: **Bhopal** 1× 2009
- National Dairy Research Institute: **Karnal** 2× 2013, 2016
- National Dairy Development Board: **Anand** 1× 2006
- Horticulture university: **Solan** (Nauni, HP) 1× 1990
- **Rice-Cuttack, farming systems-Meerut, soil science-Bhopal, agroforestry-Jhansi** 1× 2013
- **CSSRI-Karnal, CTCRI-Thiruvananthapuram, IRRI-Manila, CAZRI-Jodhpur** 1× 2013
- **Rice-Hyderabad, wheat-Karnal, maize-New Delhi, pulses-Kanpur** 1× 2013
- **Sugarcane-Lucknow, NBPGR-New Delhi, NPPTI-Hyderabad**; wheat is **Karnal**, not Kanpur (*) 1× 2014
- Plant Protection, Quarantine and Storage: **Faridabad** 1× 2013
- First **Krishi Vigyan Kendra**: **1974** (Puducherry) 1× 2017

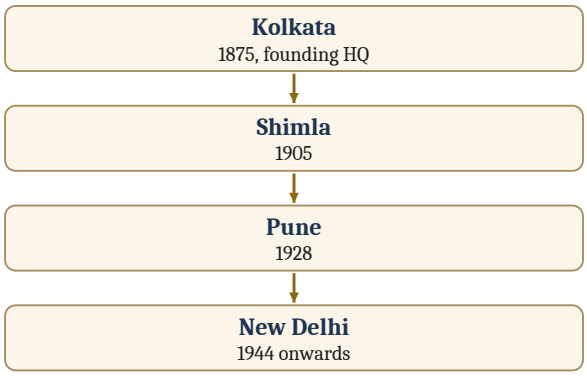
24.4 Scientific and Mining Institutes

- **NEERI-Nagpur, CFTRI-Mysuru, CPRI-Shimla, CTRI-Rajahmundry** 1× 2016
- Indian Bureau of Mines: **Nagpur** 1× 2010
- Mining institute: **Dhanbad** (ISM, now IIT-ISM) 1× 1994
- Indian Diamond Institute: **Surat** 1× 2013

QUANTUM IAS & PCS

- No Museum of Natural History: **Hyderabad** (it is in Delhi, Mysuru, Bhopal, Bhubaneswar, Sawai Madhopur) 1× 2013
- IMD HQ order: **Kolkata, Shimla, Pune, New Delhi** 1× 2018
- International Solar Alliance secretariat: **Gurugram** 1× 2019
- Central Institute of Indian Languages: **Mysuru** 1× 2019

IMD HEADQUARTERS OVER TIME



City	Institutes
Hyderabad	ICRISAT, CRIDA, NAARM, rice, NPPTI, NIRD
Karnal	NDRI, CSSRI, wheat and barley
Nagpur	NEERI, Indian Bureau of Mines, MECL, soil survey
Mysuru	CFTRI, Indian Languages
Bhopal	Soil Science, Agri Engineering, Forest Management
Cuttack	National Rice Research Institute
Jodhpur	CAZRI (arid zone)
Shimla	Potato Research
Rajahmundry	Tobacco Research

TRAP ALERT

- Same-state swaps: sugarcane **Lucknow** (not Kanpur); pulses **Kanpur** (not Lucknow); vegetables **Varanasi**; agroforestry **Jhansi** (not Banda).
- Subtropical horticulture = Lucknow; **temperate** horticulture = Srinagar.
- NDRI = **Karnal**; NDDB = **Anand**.

UP LINK

UP is the most asked state in this chapter: **Lucknow** (sugarcane, subtropical horticulture, CDRI, NBRI, CIMAP), **Kanpur** (pulses, sugar), **Varanasi** (vegetables, IRRI centre), **Jhansi** (fodder, agroforestry), **Makhdoom** (goats), **Ghaziabad** (organic farming), **Prayagraj** (eco-rehabilitation), **Agra** (potato centre).

LAST-MINUTE 10

1. Antarctic: Dakshin Gangotri, Maitri, **Bharati**; Arctic: **Himadri**
2. Pulses = **Kanpur**; sugarcane = **Lucknow**; vegetables = **Varanasi**
3. Fodder and agroforestry = **Jhansi**; goats = **Makhdoom**
4. Organic farming = **Ghaziabad**; eco-rehabilitation = **Prayagraj**
5. CRIDA, ICRISAT, NAARM = **Hyderabad**
6. NDRI = **Karnal**; NDDB = **Anand**; CAZRI = **Jodhpur**
7. Rice = **Cuttack**; potato = **Shimla**; tobacco = **Rajahmundry**
8. IBM, NEERI = **Nagpur**; CFTRI = **Mysuru**
9. IMD: Kolkata, Shimla, Pune, New Delhi
10. ISA = **Gurugram**; first KVK = **1974**

Chapter 25

Transport: Roads, Railways, Ports and Waterways

Built from 61 UPPSC questions (1990–2025). Focus: NHDP corridors, railway zones and factories, port types and port-state matching, NW-1.

ASKED MOST · REVISE FIRST

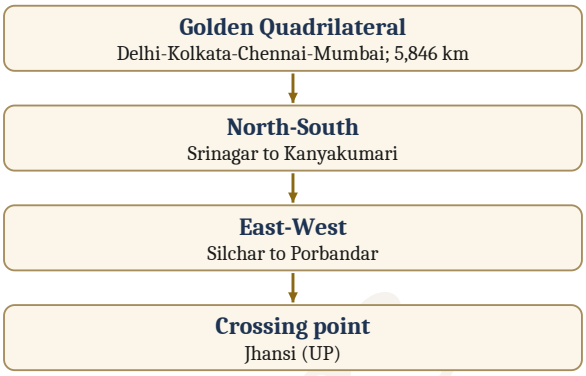
- **Port-state matching**: **Kandla/Mundra-Gujarat, Nhava Sheva-Maharashtra, Paradip-Odisha, Tuticorin-TN, Kakinada/Krishnapatnam-AP** 7× 2008, 2008, 2010, 2010, 2013, 2015, 2025
- **Port types**: deepest/landlocked **Visakhapatnam**, open sea and artificial **Chennai**, outer port **Haldia**, natural **Kochi, Visakhapatnam** 8× 1997, 2007, 2012, 2012, 2013, 2017, 2023, 2023
- **NW-1** = **Prayagraj-Haldia** on the Ganga, the longest waterway 4× 2009, 2010, 2014, 2016
- NS and EW corridors cross at **Jhansi** (UP) 2× 1999, 2001
- **Rail factories**: ICF **Perambur**, RCF **Kapurthala**, Modern Coach **Raebareli**, BLW **Varanasi** 3× 1995, 2008, 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

25.1 Roads and National Highways

- Road share of total transport: about **80%** 1× 2014
- Old **NH-4** (Mumbai-Chennai) did not touch MP, Rajasthan or UP (*) 1× 2010
- Most NH length (then): **UP**; now **Maharashtra** leads 1× 2000
- Most **state highway** length: **Maharashtra** 2× 2010, 2013
- NH length order (per key): Arunachal first, Sikkim last 1× 2004
- Old **NH-3** (Agra-Mumbai) did not touch **Bhopal** 1× 2001
- Golden Quadrilateral joins **Delhi, Kolkata, Chennai, Mumbai**; EW corridor ends at **Porbandar** 1× 2008
- **Golden Quadrilateral** = highways 1× 2004
- First six-lane expressway: **Mumbai-Pune** 2× 2001, 2003
- **PMGSY**: roads for unconnected villages 1× 2001

NHDP CORRIDORS



25.2 Railways

- First train: **Bombay-Thane, 1853** 2× 1990, 1996
- **Hyderabad**-SCR HQ is wrong (it is **Secunderabad**) 1× 2002
- Correct: **Northern-New Delhi, North Eastern-Gorakhpur**; South Eastern is **Kolkata**, not Cuttack 1× 2025
- Shortest Gorakhpur-Mumbai route: via **Prayagraj** 1× 1993
- **RCF-Kapurthala, Wheel and Axle-Bengaluru, DLW-Varanasi, ICF-Perambur** 1× 2008
- Coach-making states: **Punjab and Tamil Nadu** 1× 2008
- Third coach factory: **Raebareli** (Lalganj) 1× 1995
- Railway Staff College: **Vadodara** 1× 1991
- Bullet train: **Mumbai-Ahmedabad** 1× 2001
- First CNG train: **Rewari-Rohtak** 1× 2016
- First rail track on PPP: **Gujarat** 1× 2016
- Konkan Railway: **Roha to Mangalore** {2013}; it does **not** touch **Belgaum** 1× 2010
- State without rail in 2013: **Meghalaya** (line 2014); Sikkim is the one left now (*) 1× 2013
- Longest tunnels then: **Pir Panjal, Karbude, Nathuwadi, Berdewadi** 1× 2015
- New **Pamban** bridge: built by RVNL; India's first **vertical-lift** sea bridge (both true) 1× 2025

Zone	HQ
Northern	New Delhi
North Eastern	Gorakhpur
North Central	Prayagraj
South Eastern	Kolkata
East Coast	Bhubaneswar
South Central	Secunderabad
East Central	Hajipur
NE Frontier	Maligaon

25.3 Ports and Harbours

- Biggest port (2007 key): **Mumbai** 1× 2007

- **Kochi** is on the **Malabar** coast 1× 2013
- Landlocked harbour: **Visakhapatnam** 1× 2023
- **Chennai-deepest port** is wrong {1997}; **Kolkata-deepest** is wrong 1× 2023
- Deepest among options: **Paradip** (Visakhapatnam is the deepest landlocked) 1× 2017
- Not a natural harbour: **Chennai** {2007}; natural ports: **Kochi and Visakhapatnam** 1× 2013
- Open seaport: **Chennai** 1× 2012
- Outer port: **Haldia** (of Kolkata) 1× 2012
- Not on the east coast: **Kandla** {2008}; Odisha coast: **Paradip** 1× 2013
- **Kandla** is in **Gujarat** {1993}; on the **Gulf of Kachchh** 1× 2004
- **Paradip-Odisha, Tuticorin-TN, Kakinada-AP, Alappuzha-Kerala** 1× 2025
- **Kandla-Gujarat, Nhava Sheva-Maharashtra, Paradip-Odisha, Tuticorin-TN** 1× 2010
- **Mundra-AP** is wrong (Mundra is **Gujarat**) 1× 2015
- AP port city: **Kakinada** {2010}; **Krishnapatnam** helps **AP** 1× 2008
- 12th major port for coal: **Ennore**, near **Chennai** 2× 2004, 2010
- **Jamnagar** is not a port town (per key) 1× 2011
- Built to relieve Mumbai: **Nhava Sheva (JNPT)** 1× 2013
- No LNG terminal: **Kandla** (then) 1× 2005
- **Sethusamudram** links the **Gulf of Mannar** with **Palk Bay** 2× 2004, 2008
- **Karaikal** is not on the west coast 1× 2008

Port type	Example
Landlocked, deepest	Visakhapatnam
Oldest artificial, open sea	Chennai
Tidal	Kandla (Deendayal)
Riverine	Kolkata
Outer port	Haldia
Satellite, container	Nhava Sheva (JNPT)
Lagoon (natural)	Kochi
First corporatised	Ennore (Kamarajar)

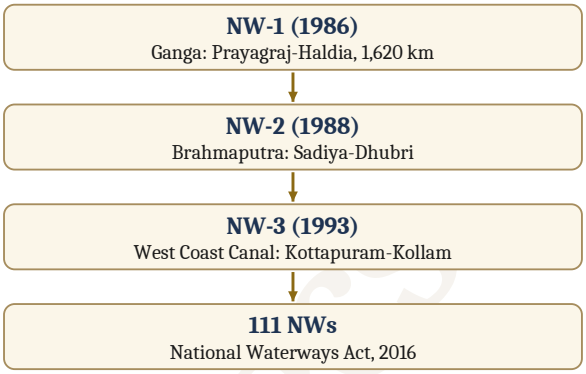
25.4 Shipyards and Ship-breaking

- **Garden Reach-WB, Hindustan Shipyard-AP, Mazagon-Maharashtra, Cochin-Kerala** 1× 2011
- Largest ship-breaking yard: **Alang** (Gujarat) 2× 2006, 2009

25.5 Waterways and Airports

- National Waterway on the Ganga: **Prayagraj to Haldia** {2010, 2016}; longest waterway 2× 2009, 2014
- National Inland Navigation Institute: **Patna** 1× 2009
- International status (24 June 2020): **Kushinagar** airport 1× 2020

NATIONAL WATERWAYS 1-3



TRAP ALERT

- **Perambur = ICF; Kapurthala = RCF.** Wheel and Axle = **Bengaluru**.
- **Visakhapatnam** = deepest landlocked; **Chennai** = artificial; **Kolkata** = shallow riverine.
- **Secunderabad**, not Hyderabad, is the SCR HQ; **Mundra** and **Kandla** are Gujarat.

UP LINK

UP links: NS-EW crossing at **Jhansi**; railway zones at **Gorakhpur** (NER) and **Prayagraj** (NCR); **Banaras Locomotive Works** (Varanasi) and **Modern Coach Factory** (Raebareli); **NW-1** starts at **Prayagraj** with a terminal at Varanasi; international airports at **Kushinagar**, Lucknow, Varanasi, Ayodhya and **Jewar** (Noida); Eastern DFC runs through UP.

LAST-MINUTE 10

1. Road = about **80%** of transport; Maharashtra = most NH and SH now
2. GQ: Delhi, Kolkata, Chennai, Mumbai; NS-EW cross at **Jhansi**
3. First train **1853 Bombay-Thane**
4. NER = **Gorakhpur**; NCR = Prayagraj; SCR = **Secunderabad**
5. ICF **Perambur**; RCF **Kapurthala**; MCF **Raebareli**
6. Konkan: **Roha-Mangalore**; bullet train Mumbai-Ahmedabad
7. **Visakhapatnam** = landlocked; **Haldia** = outer port; Chennai = artificial
8. **Kandla, Mundra** = Gujarat; Ennore = 12th major port
9. Sethusamudram = **Gulf of Mannar-Palk Bay**; Alang = ship-breaking
10. **NW-1 Prayagraj-Haldia**; Kushinagar = international airport (2020)

Chapter 26

Tourism and Miscellaneous

Built from 18 UPPSC questions (1996–2018). Focus: hill station-state matching, grasslands, temple hills, odd-one-out pairs.

ASKED MOST · REVISE FIRST

- ■ **Hill station-state matching**: **Chakrata, Auli-Uttarakhand; Kufri, Kasauli, Keylong, Khajjiar-HP; Gulmarg, Pahalgam-J&K; Haflong-Assam; Kalimpong-WB; Ooty-TN; Chikmagalur-Karnataka** 5× 1997, 2002, 2003, 2015, 2017
- ■ **Wrong-pair** traps: **Jaisalmer-City of Lakes, BARC-Karnataka, IGKV-Rajasthan, Swaminathan-geology** 4× 1996, 2004, 2005, 2016
- • Temple hills: **Biligiri Ranga** (Ranganatha, Karnataka); Tirupati = **Seshachalam** 2× 2015, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

26.1 Hill Stations and Tourist Centres

- **Chakrata-Uttarakhand, Haflong-Assam, Kalimpong-WB, Kufri-HP** 2× 2003, 2015
- **J&K-Gulmarg, HP-Kasauli, Gujarat-Udvada, TN-Point Calimere** 1× 1997
- **HP-Keylong, Uttarakhand-Auli, Karnataka-Chikmagalur, TN-Ooty** 1× 2002
- **Banni-Gujarat, Bugyal-Uttarakhand, Khajjiar-HP, Pahalgam-J&K** 1× 2017
- **Jaisalmer-City of Lakes** is wrong (that is **Udaipur**; Jaisalmer is the Golden City) 1× 2005
- Tourism campaign: **Incredible India** 1× 2010
- Not a ghost town: **Charkhari** (Mahoba, UP); ghost towns = Kuldhara, Dhanushkodi 1× 2015

Hill station	State
Chakrata, Auli, Mussoorie, Nainital	Uttarakhand
Kufri, Kasauli, Keylong, Khajjiar	Himachal Pradesh
Gulmarg, Pahalgam, Sonamarg	Jammu and Kashmir
Haflong	Assam (only hill station)
Kalimpong, Darjeeling	West Bengal
Chikmagalur, Coorg	Karnataka
Ooty, Kodaikanal, Yercaud	Tamil Nadu
Mount Abu	Rajasthan (only hill station)
Pachmarhi	Madhya Pradesh

26.2 Temple Hills

- Tirupati (Venkateswara) is on the **Seshachalam (Tirumala)** hills, AP; none of the options fitted (*) 1× 2018
- Lord Ranganatha temple: **Biligiri Ranga** Hills (Karnataka) 1× 2015

26.3 Miscellaneous

- Both power and manure: **biogas plants** 1× 2015
- **Speed Post** launched: **1986** 1× 2013
- Locusts enter India from **Pakistan** (Thar side) 1× 2010
- **BARC-Karnataka** is wrong (BARC is in **Mumbai**) 1× 2004
- Himalayan Mountaineering Institute: **Darjeeling** 1× 2008
- **Diamond Harbour, Salt Lake City**: **Kolkata** 1× 1997
- **Jangal Mahal**: **West Bengal** (Jhargram, Purulia, Bankura) 1× 2012
- **Indira Gandhi Krishi Vishwavidyalaya-Rajasthan** is wrong (it is in **Raipur, Chhattisgarh**) 1× 2016
- **M.S. Swaminathan-geology** is wrong (he was an **agricultural** scientist) 1× 1996

TRAP ALERT

- **Udaipur** = City of Lakes; **Jaisalmer** = Golden City; **Jodhpur** = Blue City; **Jaipur** = Pink City.
- **Tirupati** = Seshachalam hills (AP); **BR Hills** = Karnataka; **Shevaroy** (Yercaud) and **Javadi** = Tamil Nadu.
- **Haflong** (Assam) and **Mount Abu** (Rajasthan) are each the only hill station of their state.

UP LINK

UP tourism links: **Charkhari** (Mahoba, "Kashmir of Bundelkhand"), the **Buddhist circuit** (Sarnath, Kushinagar, Kapilvastu-Piprahwa, Shravasti, Sankisa, Kaushambi), the **Ramayana circuit** (Ayodhya, Chitrakoot, Shringverpur), **Krishna circuit** (Mathura-Vrindavan), and the **Kashi Vishwanath Corridor**.

LAST-MINUTE 10

1. Chakrata, Auli = **Uttarakhand**; Bugyal = Uttarakhand meadows
2. Kufri, Kasauli, Keylong, **Khajjiar** = **HP**
3. Gulmarg, Pahalgam = **J&K**; **Haflong** = Assam
4. Kalimpong = WB; Ooty = TN; Chikmagalur = Karnataka
5. **Banni** grassland = Gujarat; Udvada = Gujarat
6. **Udaipur** = City of Lakes, not Jaisalmer
7. Tirupati = **Seshachalam**; Ranganatha = **BR Hills**
8. **BARC** = Mumbai; HMI = **Darjeeling**
9. **Jangal Mahal** = West Bengal; IGKV = Chhattisgarh
10. Speed Post **1986**; locusts from **Pakistan**

QUANTUM IAS & PCS